

Physics 523 HW 7

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I use the notation in my Real Scalar Field & QED Notes. I usually use p for the magnitude of the momentum 3-vector, $p = |\mathbf{p}|$, so $p^2 = \mathbf{p} \cdot \mathbf{p}$. But for this homework I will use p^2 for the 4-vector dot product: $p^2 = \vec{p} \cdot \vec{p} = \vec{p}^2 = (p^0)^2 - \mathbf{p}^2$.

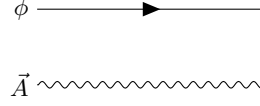


FIG. 1. Feynman lines

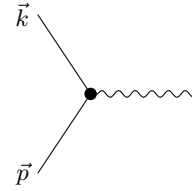


FIG. 2. The 3-way vertex with factor $ie(\pm\vec{p} \pm \vec{k})$, where $-/+$ is for particles/antiparticles.

1. SCALAR QED LAGRANGIAN

The free Lagrangian is

$$\mathcal{L}_0 = |\partial\phi|^2 - \frac{F^2}{4} - m^2|\phi|^2. \quad (1)$$

The full Lagrangian is

$$\mathcal{L} = \mathcal{L}_0 + e^2 A^2 |\phi|^2 + ieA^\mu J_\mu, \quad (2)$$

where $J_\mu = \phi \partial_\mu \phi^* - \phi^* \partial_\mu \phi$.

2. FEYNMAN RULES

I show the scalar and photon lines in Fig. 1. Their propagators are

$$\vec{A} \sim -\frac{i\eta_{\mu\nu}}{p^2 + i\epsilon}, \quad \phi \sim \frac{i}{p^2 - m^2 + i\epsilon}. \quad (3)$$

The interactions are in Figs. 2 & 3.

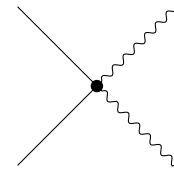


FIG. 3. The 4-way vertex with factor $2ie^2\eta_{\mu\nu}$.

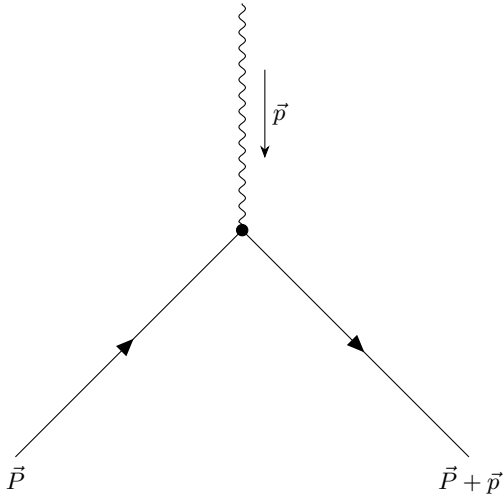


FIG. 4. Feynman diagram $i\mathcal{M}_0$. The photon is off-shell: it represents the momentum transfer with the external electromagnetic field, not an actual photon.

3. $1 \rightarrow 1$ S-MATRIX

I compute the $1 \rightarrow 1$ (no bremsstrahlung) S-matrix element to order e^3 ,

$$\mathcal{M} = \mathcal{M}_0 + \mathcal{M}_2 + \mathcal{M}_v, \quad (4)$$

where $\mathcal{M}_0$ (Fig. 4) is leading order, $\mathcal{M}_2$ is the two-point contribution, and $\mathcal{M}_v$ is the vertex correction.

3.1. $\mathcal{M}_0$

$$i\mathcal{M}_0 = ie(-\vec{P} - \vec{p} - \vec{P}) \cdot \vec{A} \quad (5)$$

If the external photon was real I would contract with its polarization. But when it represents momentum exchange with an external electromagnetic field I contract with the field.

$$\boxed{\mathcal{M}_0 = -e(\vec{p} + 2\vec{P}) \cdot \vec{A}} \quad (6)$$

3.2. $\mathcal{M}_2$

$\mathcal{M}_2$ has two diagrams (Figs. 5 & 6)

$$i\mathcal{M}_2 = \star[1] + \star[2]. \quad (7)$$

$$\star[2] = -\frac{S}{(2\pi)^4} \vec{A} \cdot \int d^4q \, ie(2\vec{P} - \vec{q}) 2ie^2 \eta_{\mu\nu} \frac{i}{(\vec{P} - \vec{q})^2 - m^2 + i\epsilon} \frac{-i\eta^{\mu\nu}}{q^2 - \lambda^2 + i\epsilon} \quad (8)$$

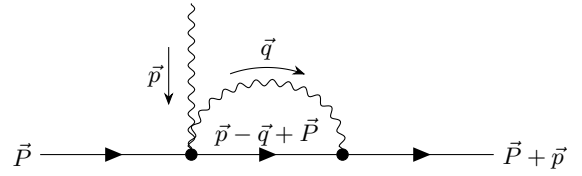


FIG. 5. $\star[1]$

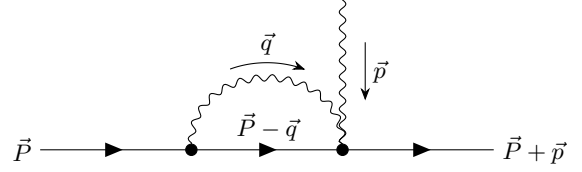


FIG. 6. $\star[2]$

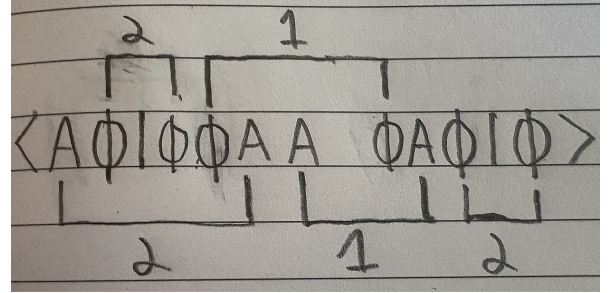


FIG. 7. Symmetry factor combinatorics for $\star[1]$. By symmetry the symmetry factor for $\star[2]$ must be the same.

S is the symmetry factor (Fig. 7) and λ is a fictitious photon mass to regulate IR divergences.

$$S = \frac{2^3}{(2!)^3} = 1 \quad (9)$$

$$\star[2] = \frac{e^3 I}{8\pi^4} \eta_{\mu\nu} \eta^{\mu\nu} \quad (10)$$

$$I = \vec{A} \cdot \int d^4 q \frac{2\vec{P} - \vec{q}}{[(\vec{P} - \vec{q})^2 - m^2 + i\epsilon](q^2 - \lambda^2 + i\epsilon)} \quad (11)$$

$$\eta_{\mu\nu}\eta^{\mu\nu} = 4 \quad (12)$$

$$\star[2] = \frac{e^3 I}{2\pi^4} \quad (13)$$

3.2.1. Feynman Parameters

$$B = q^2 - \lambda^2 + i\epsilon, \quad C = (\vec{P} - \vec{q})^2 - m^2 + i\epsilon, \quad D = B + (C - B)x \quad (14)$$

$$I = \vec{A} \cdot \int d^4 q \frac{2\vec{P} - \vec{q}}{BC} = \vec{A} \cdot \int_0^1 dx \int d^4 q \frac{2\vec{P} - \vec{q}}{D^2} \quad (15)$$

$$D = q^2 - 2x\vec{P} \cdot \vec{q} - \lambda^2 + (P^2 - m^2 + \lambda^2)x + i\epsilon \quad (16)$$

$$P^2 = m^2 \quad (17)$$

$$D = q^2 - 2x\vec{P} \cdot \vec{q} - (1 - x)\lambda^2 + i\epsilon \quad (18)$$

3.2.2. Complete the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + c - \frac{b^2}{4a} \quad (19)$$

$$x \rightarrow \vec{q}, \quad a \rightarrow 1, \quad b \rightarrow -2x\vec{P}, \quad c \rightarrow (x - 1)\lambda^2 + i\epsilon \quad (20)$$

$$D = (\vec{q} - x\vec{P})^2 - \kappa + i\epsilon \quad (21)$$

$$\kappa = (1 - x)\lambda^2 + x^2 m^2 \quad (22)$$

$$I = \vec{A} \cdot \int_0^1 dx \int d^4 k \frac{2\vec{P} - \vec{k}}{[(\vec{k} - x\vec{P})^2 - \kappa + i\epsilon]^2} \quad (23)$$

3.2.3. Shift Integrand

$$\vec{q} = \vec{k} - x\vec{P}, \quad \vec{k} = \vec{q} + x\vec{P}, \quad d^4 q = d^4 k \quad (24)$$

$$I = \vec{A} \cdot \int_0^1 dx \int d^4 q \frac{(2 - x)\vec{P} - \not{q}}{(q^2 - \kappa + i\epsilon)^2} \quad (25)$$

The odd term integrates to 0.

$$I = \vec{A} \cdot \vec{P} \int_0^1 dx (2 - x) \int \frac{d^4 q}{(q^2 - \kappa + i\epsilon)^2} \quad (26)$$

$$I_1 = \int \frac{d^4 q}{(q^2 - \kappa + i\epsilon)^2} \quad (27)$$

$$I = \vec{A} \cdot \vec{P} \int_0^1 dx (2 - x) I_1 \quad (28)$$

$$\star[2] = \frac{e^3}{2\pi^4} \vec{A} \cdot \vec{P} \int_0^1 dx (2 - x) I_1 \quad (29)$$

3.2.4. UV Divergence

$$I_1 \sim \int_1^\Lambda \frac{dq}{q} = \ln \Lambda, \quad \Lambda \rightarrow \infty \quad (30)$$

$\star[2]$ has a logarithmic UV divergence.

3.2.5. Wick Rotation

Before testing for IR divergences I need to Wick rotate. Otherwise the momentum integral diverges and it becomes impossible to see if there is an IR divergence; this happened during my first attempt.

$$\star[2] = \frac{e^3}{2\pi^4} \vec{A} \cdot \vec{P} \int_0^1 dx (2 - x) I_1 \quad (31)$$

$$I_1 = \int \frac{d^4 q}{(q^2 - \kappa + i\epsilon)^2} \quad (32)$$

In my Real Scalar Field Notes I derived

$$\int \frac{d^4 p}{(p^2 - \Delta + i\epsilon)^n} = i \int \frac{d^4 p_E}{(-p_E^2 - \Delta)^n}, \quad (33)$$

where $p_E^2 = \delta_{\alpha\beta} p^\alpha p^\beta$.

$$I_1 = i \int \frac{d^4 q_E}{(q_E^2 + \kappa)^2} \quad (34)$$

$$\star[2] = \frac{ie^3}{2\pi^4} \vec{A} \cdot \vec{P} \int_0^1 dx (2 - x) \int \frac{d^4 q_E}{(q_E^2 + \kappa)^2} \quad (35)$$

3.2.6. IR Divergence

$$\star[2] \sim \int_0^1 dx (2-x) \int \frac{d^4 q_E}{(q_E^2 + \kappa)^2} \quad (36)$$

$$\kappa = (1-x)\lambda^2 + x^2 m^2 \quad (37)$$

To test for IR divergences I set $\lambda = 0$.

$$\star[2] \sim \int_0^1 dx (2-x) \int \frac{d^4 q_E}{(q_E^2 + x^2 m^2)^2} \quad (38)$$

$$\star[2] \sim \int_0^1 dx (2-x) \int_0^\infty dq \frac{q^3}{(q^2 + x^2 m^2)^2} \quad (39)$$

The IR divergence can only appear around $q = x = 0$.

$$\star[2] \sim \int_0^\delta dx (2-x) \int_0^{m\delta} dq \frac{q^3}{(q^2 + x^2 m^2)^2}, \quad \delta \rightarrow 0_+ \quad (40)$$

Where δ is dimensionless. Since I am power counting I set $m = 1$.

$$\star[2] \sim \int_0^\delta dx (2-x) \int_0^\delta dq \frac{q^3}{(q^2 + x^2)^2} \quad (41)$$

$$\star[2] \sim 2 \int_0^\delta dx \int_0^\delta dq \frac{q^3}{(q^2 + x^2)^2} \quad (42)$$

$$\star[2] \sim \int_0^\delta dx \int_0^\delta dq \frac{q^3}{(q^2 + x^2)^2} \quad (43)$$

$$q = xt, \quad t = q/x, \quad t(0) = 0, \quad t(\delta) = \delta/x, \quad dq = xdt \quad (44)$$

$$\star[2] \sim \int_0^\delta dx \int_0^{\delta/x} dt \frac{t^3}{(t^2 + 1)^2} \quad (45)$$

$$z = x/\delta, \quad z(0) = 0, \quad z(\delta) = 1, \quad dx = \delta dz \quad (46)$$

$$\star[2] \sim \delta \int_0^1 dz \int_0^{1/z} dt \frac{t^3}{(t^2 + 1)^2} \quad (47)$$

$$2 \int_0^a dx \frac{x^3}{(1+x^2)^2} = \ln(1+a^2) - \frac{a^2}{1+a^2} \quad (48)$$

$$\star[2] \sim \frac{\delta}{2} \int_0^1 dz \left[\ln \left(1 + \frac{1}{z^2} \right) - \frac{1}{1+z^2} \right] \quad (49)$$

$$\int_0^1 dx \ln \left(1 + \frac{1}{x^2} \right) = \frac{\pi}{2} + \ln 2, \quad \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4} \quad (50)$$

$$\star[2] \sim \lim_{\delta \rightarrow 0_+} \delta = 0 \quad (51)$$

There is no IR divergence.

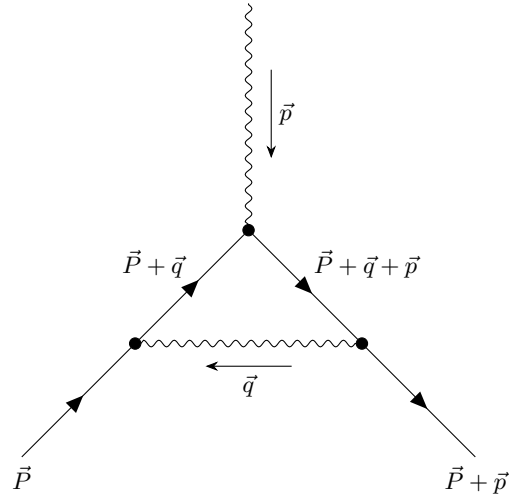


FIG. 8. $i\mathcal{M}_v$

3.2.7. Final

By symmetry $\star[1]$, like $\star[2]$, also has a UV divergence, but does not have an IR divergence. So $\mathcal{M}_2$ (Eq. 7) is UV, not IR, divergent. In this problem I only care about IR divergences, which is why I did not include counterterms. All I need is

$$\boxed{\mathcal{M}_2 = e^3 \mathcal{U}}, \quad (52)$$

where $\mathcal{U}$ means IR convergent: I use this label for both finite and UV divergent terms.

3.3. $\mathcal{M}_v$

The vertex correction diagram is Fig. 8.

$$i\mathcal{M}_v = \frac{S}{(2\pi)^4} \int d^4 q ie[-P^\mu - (P^\mu + q^\mu)]ie \vec{A} \cdot [-(\vec{P} + \vec{q}) - (\vec{P} + \vec{q} + \vec{p})]ie[-(P^\nu + q^\nu + p^\nu) - (p^\nu + P^\nu)] \frac{i}{(\vec{P} + \vec{q})^2 - m^2 + i\epsilon} \frac{i}{(\vec{P} + \vec{q} + \vec{p})^2 - m^2 + i\epsilon} \frac{-i\eta_{\mu\nu}}{q^2 - \lambda^2 + i\epsilon} \quad (53)$$

S , Fig. 9, is the symmetry factor and λ is a fictitious photon mass to regulate IR divergences.

$$S = \frac{6 \times 4 \times 2}{3!(2!)^3} = 1 \quad (54)$$

$$\mathcal{M}_v = \frac{ie^3}{(2\pi)^4} \vec{A} \cdot \int d^4 q \frac{2\vec{P} + 2\vec{q} + \vec{p}}{q^2 - \lambda^2 + i\epsilon} \frac{2P^\mu + q^\mu}{(\vec{P} + \vec{q})^2 - m^2 + i\epsilon} \frac{(2P^\nu + 2p^\nu + q^\nu)\eta_{\mu\nu}}{(\vec{P} + \vec{q} + \vec{p})^2 - m^2 + i\epsilon} \quad (55)$$

$$D = [\vec{q} + y\vec{p} + (1-z)\vec{P}]^2 - \Delta + i\epsilon \quad (73)$$

$$\Delta = z\lambda^2 + [y\vec{p} + (1-z)\vec{P}]^2 \quad (74)$$

$$\Delta = z\lambda^2 + y^2 p^2 + (1-z)^2 P^2 + 2y(1-z)\vec{p} \cdot \vec{P} \quad (75)$$

$$\Delta = z\lambda^2 + (1-z)^2 m^2 + y(y+z-1)p^2 \quad (76)$$

$$\Delta = z\lambda^2 + (1-z)^2 m^2 - xyp^2 \quad (77)$$

3.3.3. Shift Integrand

$$\begin{aligned} \mathcal{M}_V &= \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \\ &\int d^4 k \vec{A} \cdot (\vec{p} + 2\vec{k} + 2\vec{P}) \frac{2(\vec{p} + 2\vec{P}) \cdot \vec{k} + k^2 - 2p^2 + 4m^2}{[\vec{k} + y\vec{p} + (1-z)\vec{P}]^2 - \Delta + i\epsilon} \end{aligned} \quad (78)$$

$$\vec{q} = \vec{k} + y\vec{p} + (1-z)\vec{P}, \quad \vec{k} = \vec{q} - y\vec{p} - (1-z)\vec{P}, \quad d^4 q = d^4 k \quad (79)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)N}{(q^2 - \Delta + i\epsilon)^3} \quad (80)$$

$$\begin{aligned} N &= \vec{A} \cdot [2\vec{q} + (1-2y)\vec{p} + 2z\vec{P}] [4m^2 + (z^2 + 2z - 3)P^2 + q^2 \\ &+ (y^2 - 2 - 2y)p^2 - 2(1+y-z+yz)\vec{P} \cdot \vec{p} + 2(1-y)\vec{q} \cdot \vec{p} \\ &+ 2(1+z)\vec{q} \cdot \vec{P}] \end{aligned} \quad (81)$$

$$P^2 = m^2, \quad 2\vec{p} \cdot \vec{P} = -p^2 \quad (82)$$

$$\begin{aligned} N &= \vec{A} \cdot [2\vec{q} + (1-2y)\vec{p} + 2z\vec{P}] [(1+z)^2 m^2 + q^2 \\ &- (1+y+z-y^2-yz)p^2 + 2(1-y)\vec{q} \cdot \vec{p} + 2(1+z)\vec{q} \cdot \vec{P}] \end{aligned} \quad (83)$$

$$N = \vec{A} \cdot (2\vec{q} + \vec{K})(\sigma + q^2 + 2\vec{q} \cdot \vec{V}) \quad (84)$$

$$\begin{aligned} \vec{K} &= (1-2y)\vec{p} + 2z\vec{P}, \quad \vec{V} = (1-y)\vec{p} + (1+z)\vec{P} \\ \sigma &= (1+z)^2 m^2 - (1+y+z-y^2-yz)p^2 \end{aligned} \quad (85)$$

$$N = 2[\sigma \vec{A} + q^2 \vec{A} + (\vec{A} \cdot \vec{K})\vec{V}] \cdot \vec{q} + \mathcal{N} \quad (86)$$

$$\mathcal{N} = 4(\vec{q} \cdot \vec{V})(\vec{q} \cdot \vec{A}) + (\sigma + q^2)\vec{A} \cdot \vec{K} \quad (87)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{N}{(q^2 - \Delta + i\epsilon)^3} \quad (88)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{2[\sigma \vec{A} + q^2 \vec{A} + (\vec{A} \cdot \vec{K})\vec{V}] \cdot \vec{q} + \mathcal{N}}{(q^2 - \Delta + i\epsilon)^3} \quad (89)$$

Odd terms integrate to 0.

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \int d^4 q \frac{\delta(x+y+z-1)\mathcal{N}}{(q^2 - \Delta + i\epsilon)^3} \quad (90)$$

$$\mathcal{N} = 4V_\mu A_\nu q^\mu q^\nu + (\sigma + q^2)\vec{A} \cdot \vec{K} \quad (91)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{\mathcal{N}}{(q^2 - \Delta + i\epsilon)^3} \quad (92)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{4V_\mu A_\nu q^\mu q^\nu + (\sigma + q^2)\vec{A} \cdot \vec{K}}{(q^2 - \Delta + i\epsilon)^3} \quad (93)$$

I use the identity (Schwartz Eq. B.51)

$$\int d^4 p \frac{p^\mu p^\nu}{(p^2 + a)^n} = \eta^{\mu\nu} \chi \int d^4 p \frac{p^2}{(p^2 + a)^n} \quad (94)$$

where $\chi = 1/4$ and 4 is the number of dimensions. I use χ to track factors of 4, so that I can replace $4 \rightarrow d$ when I use DimReg.

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{4V_\mu A_\nu \eta^{\mu\nu} \chi q^2 + (\sigma + q^2)\vec{A} \cdot \vec{K}}{(q^2 - \Delta + i\epsilon)^3} \quad (95)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{4\vec{V} \cdot \vec{A} \chi q^2 + (\sigma + q^2)\vec{A} \cdot \vec{K}}{(q^2 - \Delta + i\epsilon)^3} \quad (96)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4 q \frac{4\vec{V} \chi q^2 + (\sigma + q^2)\vec{K}}{(q^2 - \Delta + i\epsilon)^3} \quad (97)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \vec{A} \cdot \int_0^1 dx dy dz \int d^4q \frac{\delta(x+y+z-1) \vec{\mathcal{N}}}{(q^2 - \Delta + i\epsilon)^3} \quad (98)$$

$$\vec{\mathcal{N}} = 4\vec{V}\chi q^2 + (\sigma + q^2)\vec{K} \quad (99)$$

$$\vec{\mathcal{N}} = \sigma\vec{K} + \vec{W}q^2 \quad (100)$$

$$\vec{W} = 4\vec{V}\chi + \vec{K} \quad (101)$$

$$\vec{K} = (1-2y)\vec{p} + 2z\vec{P}, \quad \vec{V} = (1-y)\vec{p} + (1+z)\vec{P} \quad (102)$$

$$\vec{W} = [4\chi(1-y) + 1 - 2y]\vec{p} + 2[2\chi(1+z) + z]\vec{P} \quad (103)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4q \frac{\vec{\mathcal{N}}}{(q^2 - \Delta + i\epsilon)^3} \quad (104)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \int d^4q \frac{\sigma\vec{K} + \vec{W}q^2}{(q^2 - \Delta + i\epsilon)^3} \quad (105)$$

$$\mathbb{I} = \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^3}, \quad \Xi = \int d^4q \frac{q^2}{(q^2 - \Delta + i\epsilon)^3} \quad (106)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) (\sigma\vec{K}\mathbb{I} + \vec{W}\Xi) \quad (107)$$

$$\mathcal{J} = \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (108)$$

$$\Xi = \mathcal{J} + (\Delta - i\epsilon)\mathbb{I} \quad (109)$$

I take the $\epsilon \rightarrow 0$ limit in the numerator.

$$\Xi = \mathcal{J} + \Delta\mathbb{I} \quad (110)$$

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) [(\sigma\vec{K} + \vec{W}\Delta)\mathbb{I} + \vec{W}\mathcal{J}] \quad (111)$$

3.3.4. UV Divergence

$$\mathcal{J} \sim \int_1^\Lambda \frac{dq}{q} = \ln\Lambda, \quad \Lambda \rightarrow \infty \quad (112)$$

$$\mathbb{I} \sim \int_1^\Lambda \frac{dq}{q^3} = \frac{1}{2} \left(1 - \frac{1}{\Lambda^2}\right) = \frac{1}{2} \quad (113)$$

$\mathcal{J}$ is UV divergent, but $\mathbb{I}$ is UV finite.

3.3.5. Wick Rotation

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) [(\sigma\vec{K} + \vec{W}\Delta)\mathbb{I} + \vec{W}\mathcal{J}] \quad (114)$$

$$\mathbb{I} = \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^3}, \quad \mathcal{J} = \int \frac{d^4q}{(q^2 - \Delta + i\epsilon)^2} \quad (115)$$

$$\int \frac{d^4p}{(p^2 - \Delta + i\epsilon)^n} = i \int \frac{d^4p_E}{(-p_E^2 - \Delta)^n}, \quad (116)$$

$$\mathbb{I} = -i \int \frac{d^4q_E}{(q_E^2 + \Delta)^3}, \quad \mathcal{J} = i \int \frac{d^4q_E}{(q_E^2 + \Delta)^2} \quad (117)$$

3.3.6. IR Divergence

$$\mathcal{M}_V = \frac{2ie^3}{(2\pi)^4} (T_1 + T_2 + T_3) \quad (118)$$

$$\begin{aligned} T_1 &= \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \sigma\vec{K}\mathbb{I} \\ T_2 &= \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \vec{W}\Delta\mathbb{I} \\ T_3 &= \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \vec{W}\mathcal{J} \end{aligned} \quad (119)$$

$$\mathbb{I} = -i \int \frac{d^4q_E}{(q_E^2 + \Delta)^3}, \quad \mathcal{J} = i \int \frac{d^4q_E}{(q_E^2 + \Delta)^2} \quad (120)$$

$$\begin{aligned} \vec{K} &= (1-2y)\vec{p} + 2z\vec{P}, \quad \chi = 1/4, \\ \vec{W} &= [4\chi(1-y) + 1 - 2y]\vec{p} + 2[2\chi(1+z) + z]\vec{P} \\ \sigma &= (1+z)^2m^2 - (1+y+z-y^2-yz)p^2 \end{aligned} \quad (121)$$

$$\Delta = z\lambda^2 + (1-z)^2m^2 - xyp^2 \quad (122)$$

To test for IR divergences I set $\lambda = 0$.

$$T_1 \sim \vec{A} \cdot \int_0^1 dx dy dz \int d^4q_E \frac{\delta(x+y+z-1) \sigma\vec{K}}{[q_E^2 + (1-z)^2m^2 - xyp^2]^3} \quad (123)$$

$$T_1 \sim \vec{A} \cdot \int_0^1 dx dy dz \int_0^{m\delta} dq \frac{q^3 \delta(x+y+z-1) \sigma\vec{K}}{[q^2 + (1-z)^2m^2 - xyp^2]^3}, \quad \delta \rightarrow 0_+ \quad (124)$$

An IR divergence can only occur when $q = x = y = 0$ which implies, because of the delta function, that $z = 1$. I need m in the upper limit for unit consistency because δ is dimensionless.

$$T_1 \sim \vec{A} \cdot \int_0^\delta dx dy \int_0^1 dz \delta(x+y+z-1) \int_0^{m\delta} dq \frac{q^3 \sigma \vec{K}}{[q^2 + (1-z)^2 m^2 - xyp^2]^3} \quad (125)$$

$$T_1 \sim \int_0^\delta dx dy \int_0^1 dz \delta(x+y+z-1) [(1+z)^2 m^2 - (1+y+z-y^2-yz)p^2] \vec{A} \cdot [(1-2y)\vec{p} + 2z\vec{P}] \int_0^{m\delta} dq \frac{q^3}{[q^2 + (1-z)^2 m^2 - xyp^2]^3} \quad (126)$$

When power counting I can set $\vec{A} \cdot \vec{p} = \vec{A} \cdot \vec{P} = m = p^2 = 1$.

$$T_1 \sim \int_0^\delta dx dy \int_0^1 dz \delta(x+y+z-1) (1-2y+2z) \int_0^\delta dq \frac{q^3 (z-y+y^2+z^2+yz)}{[q^2 + (1-z)^2 - xy]^3} \quad (127)$$

$$T_1 \sim \int_0^\delta dx dy dq (3-4y-2x) \frac{q^3 (2-3x-3y+x^2+y^2+xy)}{(q^2+x^2+y^2+xy)^3} \quad (128)$$

$$T_1 \sim 6 \int_0^\delta dx dy dq \frac{q^3}{(q^2+x^2+y^2+xy)^3} \quad (129)$$

$$T_1 \sim \int_0^\delta dx dy \int_0^\delta dq \left(\frac{q}{q^2+x^2+y^2+xy} \right)^3 \quad (130)$$

$$\int_0^a dx \left(\frac{x}{x^2+b} \right)^3 = \frac{a^4}{4b(a^2+b)^2}, \quad a > 0, \quad b > 0 \quad (131)$$

$$T_1 \sim \delta^4 \int_0^\delta \frac{dx dy}{(x^2+y^2+xy)(\delta^2+x^2+y^2+xy)^2} \quad (132)$$

$$T_1 \sim \delta^4 \int_0^\delta dx \int_0^\delta dy \frac{1}{(x^2+y^2+xy)(\delta^2+x^2+y^2+xy)^2} \quad (133)$$

$$\begin{aligned} x &= u\delta, \quad u = x/\delta, \quad u(0) = 0, \quad u(\delta) = 1, \quad dx = \delta du \\ y &= v\delta, \quad v = y/\delta, \quad v(0) = 0, \quad v(\delta) = 1, \quad dy = \delta dv \end{aligned} \quad (134)$$

$$T_1 \sim \delta^6 \int_0^1 du \int_0^1 dv \frac{1}{(u^2\delta^2 + v^2\delta^2 + uv\delta^2)(\delta^2 + u^2\delta^2 + v^2\delta^2 + uv\delta^2)^2} \quad (135)$$

$$T_1 \sim \frac{\delta^6}{\delta^6} \int_0^1 \frac{du dv}{(u^2 + v^2 + uv)(1 + u^2 + v^2 + uv)^2} \quad (136)$$

$$T_1 \sim \int_0^1 \frac{du dv}{(u^2 + v^2 + uv)(1 + u^2 + v^2 + uv)^2} = \infty \quad (137)$$

$$T_1 \sim \infty \quad (138)$$

Even though δ cancels out the remaining integral diverges, so T_1 has an IR divergence.

$$T_2 \sim \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \vec{W} \Delta \int \frac{d^4 q_E}{(q_E^2 + \Delta)^3} \quad (139)$$

$$T_2 \sim \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \vec{W} \Delta \int_0^{m\delta} dq \left(\frac{q}{q^2 + \Delta} \right)^3 \quad (140)$$

$$\begin{aligned} T_2 &\sim \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) [z\lambda^2 + (1-z)^2 m^2 - xyp^2] \\ &\quad [4\chi(1-y) + 1 - 2y] \vec{p} + 2[2\chi(1+z) + z] \vec{P} \\ &\quad \int_0^{m\delta} dq \left(\frac{q}{q^2 + z\lambda^2 + (1-z)^2 m^2 - xyp^2} \right)^3 \end{aligned} \quad (141)$$

I am testing for IR divergences, so I can use $\chi = 1/4$, because an IR test does not involve DimReg.

$$\begin{aligned} T_2 &\sim \int_0^1 dx dy dz \delta(x+y+z-1) [z\lambda^2 + (1-z)^2 m^2 - xyp^2] \\ &\quad \vec{A} \cdot [(2-3y)\vec{p} + (1+3z)\vec{P}] \\ &\quad \int_0^{m\delta} dq \left[\frac{q}{q^2 + z\lambda^2 + (1-z)^2 m^2 - xyp^2} \right]^3 \end{aligned} \quad (142)$$

$$\lambda \rightarrow 0 \quad (143)$$

$$\begin{aligned} T_2 &\sim \int_0^1 dx dy dz \delta(x+y+z-1) [(1-z)^2 m^2 - xyp^2] \\ &\quad [(2-3y)\vec{A} \cdot \vec{p} + (1+3z)\vec{A} \cdot \vec{P}] \\ &\quad \int_0^{m\delta} dq \left[\frac{q}{q^2 + (1-z)^2 m^2 - xyp^2} \right]^3 \end{aligned} \quad (144)$$

$$m = p^2 = \vec{A} \cdot \vec{p} = \vec{A} \cdot \vec{P} = 1 \quad (145)$$

$$T_2 \sim 3 \int_0^1 dx dy dz \delta(x+y+z-1)[(1-z)^2 - xy] \\ (1-y+z) \int_0^\delta dq \left[\frac{q}{q^2 + (1-z)^2 - xy} \right]^3 \quad (146)$$

$$T_2 \sim \int_0^\delta dx dy \int_0^1 dz \delta(x+y+z-1)[(1-z)^2 - xy] \\ (1-y+z) \int_0^\delta dq \left[\frac{q}{q^2 + (1-z)^2 - xy} \right]^3 \quad (147)$$

$$T_2 \sim \int_0^\delta dx dy [(x+y)^2 - xy](2-2y-x) \\ \int_0^\delta dq \left[\frac{q}{q^2 + (x+y)^2 - xy} \right]^3 \quad (148)$$

$$T_2 \sim \int_0^\delta dx dy (x^2 + y^2 + xy)(2-2y-x) \\ \int_0^\delta dq \left(\frac{q}{q^2 + x^2 + y^2 + xy} \right)^3 \quad (149)$$

$$T_2 \sim 2 \int_0^\delta dx dy (x^2 + y^2 + xy) \\ \int_0^\delta dq \left(\frac{q}{q^2 + x^2 + y^2 + xy} \right)^3 \quad (150)$$

$$T_2 \sim \int_0^\delta dx dy (x^2 + y^2 + xy) \\ \int_0^\delta dq \left(\frac{q}{q^2 + x^2 + y^2 + xy} \right)^3 \quad (151)$$

$$\int_0^a dx \left(\frac{x}{x^2 + b} \right)^3 = \frac{a^4}{4b(a^2 + b)^2}, \quad a > 0, \quad b > 0 \quad (152)$$

$$T_2 \sim \delta^4 \int_0^\delta dx dy \frac{x^2 + y^2 + xy}{(x^2 + y^2 + xy)(\delta^2 + x^2 + y^2 + xy)^2} \quad (153)$$

$$T_2 \sim \delta^4 \int_0^\delta \frac{dx dy}{(\delta^2 + x^2 + y^2 + xy)^2} \quad (154)$$

$$x = u\delta, \quad u = x/\delta, \quad u(0) = 0, \quad u(\delta) = 1, \quad dx = \delta du \\ y = v\delta, \quad v = y/\delta, \quad v(0) = 0, \quad v(\delta) = 1, \quad dy = \delta dv \quad (155)$$

$$T_2 \sim \delta^6 \int_0^1 \frac{du dv}{(\delta^2 + u^2\delta^2 + v^2\delta^2 + \delta^2 uv)^2} \quad (156)$$

$$T_2 \sim \frac{\delta^6}{\delta^4} \int_0^1 \frac{du dv}{(1 + u^2 + v^2 + uv)^2} \quad (157)$$

$$T_2 \sim \delta^2 \int_0^1 \frac{du dv}{(1 + u^2 + v^2 + uv)^2} \quad (158)$$

$$\int_0^1 \frac{dx dy}{(1 + x^2 + y^2 + xy)^2} = \frac{2}{\sqrt{7}} \operatorname{acot}\left(\frac{5}{\sqrt{7}}\right) \approx 0.37 \quad (159)$$

$$T_2 \sim \delta^2 \sim 0, \quad \delta \rightarrow 0_+ \quad (160)$$

T_2 is IR convergent.

$$T_3 \sim \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \\ [[4\chi(1-y)+1-2y]\vec{p}+2[2\chi(1+z)+z]\vec{P}] \int_0^{m\delta} dq \frac{q^3}{(q^2 + \Delta)^2} \quad (161)$$

$$\delta \rightarrow 0_+, \quad \chi = 1/4 \quad (162)$$

$$T_3 \sim \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \\ [(2-3y)\vec{p} + (1+3z)\vec{P}] \int_0^{m\delta} dq \frac{q^3}{(q^2 + \Delta)^2} \quad (163)$$

$$\Delta = z\lambda^2 + (1-z)^2 m^2 - xyp^2 \quad (164)$$

$$T_3 \sim \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1)[(2-3y)\vec{p} + (1+3z)\vec{P}] \\ \int_0^{m\delta} dq \frac{q^3}{[q^2 + z\lambda^2 + (1-z)^2 m^2 - xyp^2]^2} \quad (165)$$

$$\lambda \rightarrow 0 \quad (166)$$

$$T_3 \sim \int_0^1 dx dy dz \delta(x+y+z-1)[(2-3y)\vec{A} \cdot \vec{p} + (1+3z)\vec{A} \cdot \vec{P}] \\ \int_0^{m\delta} dq \frac{q^3}{[q^2 + (1-z)^2 m^2 - xyp^2]^2} \quad (167)$$

$$m = p^2 = \vec{A} \cdot \vec{p} = \vec{A} \cdot \vec{P} = 1 \quad (168)$$

$$T_3 \sim 3 \int_0^1 dx dy dz \delta(x+y+z-1)(1-y+z) \\ \int_0^\delta dq \frac{q^3}{[q^2 + (1-z)^2 - xy]^2} \quad (169)$$

$$T_3 \sim \int_0^1 dx dy dz \delta(x+y+z-1)(1-y+z) \int_0^\delta dq \frac{q^3}{[q^2 + (1-z)^2 - xy]^2} \quad (170)$$

$$T_3 \sim \int_0^\delta dx dy \int_0^1 dz \delta(x+y+z-1)(1-y+z) \int_0^\delta dq \frac{q^3}{[q^2 + (1-z)^2 - xy]^2} \quad (171)$$

$$T_3 \sim \int_0^\delta dx dy \int_0^\delta dq \frac{q^3(2-x-2y)}{(q^2 + x^2 + y^2 + xy)^2} \quad (172)$$

$$T_3 \sim 2 \int_0^\delta dx dy \int_0^\delta dq \frac{q^3}{(q^2 + x^2 + y^2 + xy)^2} \quad (173)$$

$$T_3 \sim \int_0^\delta dx dy \int_0^\delta dq \frac{q^3}{(q^2 + x^2 + y^2 + xy)^2} \quad (174)$$

$$2 \int_0^a dx \frac{x^3}{(x^2 + b)^2} = \ln \left(\frac{a^2 + b}{b} \right) - \frac{a^2}{a^2 + b}, \quad a, b > 0 \quad (175)$$

$$T_3 \sim \int_0^\delta dx dy \ln \left(\frac{\delta^2 + x^2 + y^2 + xy}{x^2 + y^2 + xy} \right) - \delta^2 \int_0^\delta \frac{dx dy}{\delta^2 + x^2 + y^2 + xy} \quad (176)$$

$$\begin{aligned} x &= u\delta, \quad u = x/\delta, \quad u(0) = 0, \quad u(\delta) = 1, \quad dx = \delta du \\ y &= v\delta, \quad v = y/\delta, \quad v(0) = 0, \quad v(\delta) = 1, \quad dy = \delta dv \end{aligned} \quad (177)$$

$$T_3 \sim \delta^2 \int_0^1 du dv \ln \left(\frac{1 + u^2 + v^2 + uv}{u^2 + v^2 + uv} \right) - \delta^2 \int_0^1 \frac{du dv}{1 + u^2 + v^2 + uv} \quad (178)$$

$$\int_0^1 dx dy \ln \left(\frac{1 + x^2 + y^2 + xy}{x^2 + y^2 + xy} \right) \approx 1.04, \quad \int_0^1 \frac{dx dy}{1 + x^2 + y^2 + xy} \approx 0.58 \quad (179)$$

$$T_3 \sim \delta^2 C = 0, \quad \delta \rightarrow 0_+ \quad (180)$$

C is a constant. T_3 is IR convergent. Only T_1 has an IR divergence: it is the only term I need to track in detail.

$$\mathcal{M}_v = \frac{2ie^3}{(2\pi)^4} T_1 + \frac{2ie^3}{(2\pi)^4} (T_2 + T_3) \quad (181)$$

I group the IR convergent terms together (their details don't matter for this problem).

$$\mathcal{U} = i \frac{T_2 + T_3}{8\pi^4} \quad (182)$$

$$\mathcal{M}_v = \frac{ie^3}{8\pi^4} T_1 + e^3 \mathcal{U} \quad (183)$$

3.3.7. Isolate IR Divergence

$$T_1 = \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \sigma \vec{K} \mathbb{I} \quad (184)$$

$$\begin{aligned} \sigma &= (1+z)^2 m^2 - (1+y+z-y^2-yz)p^2, \\ \vec{K} &= (1-2y)\vec{p} + 2z\vec{P}, \quad \Delta = z\lambda^2 + (1-z)^2 m^2 - xyp^2 \end{aligned} \quad (185)$$

$$\mathbb{I} = -i \int \frac{d^4 q_E}{(q_E^2 + \Delta)^3}, \quad q_E^2 = (q^0)^2 + (q^1)^2 + (q^2)^2 + (q^3)^2 \quad (186)$$

In Sec. 3.3.6 I showed that the IR divergence occurs near $x = y = q^\beta = \lambda = 0$. I need to isolate this region. Using spherical symmetry and my Real Scalar Field Notes I write

$$i\mathbb{I} = \int d\Omega \int_0^\infty dr \left(\frac{r}{r^2 + \Delta} \right)^3 = \Omega \int_0^\infty dr \left(\frac{r}{r^2 + \Delta} \right)^3, \quad (187)$$

where $\Omega = 2\pi^2$ is the surface volume of a unit 3-sphere.

$$i\mathbb{I} = 2\pi^2 \int_0^\infty dr \left(\frac{r}{r^2 + \Delta} \right)^3 \quad (188)$$

$$\int_0^\infty dr \left(\frac{r}{r^2 + \Delta} \right)^3 = \frac{1}{4\Delta}, \quad \Delta > 0 \quad (189)$$

This integral diverges for $\Delta \leq 0$. And in Eq. 185 it appears that some values of xyp^2 can turn Δ negative. But we are working in the IR divergent region near $x = y = 0$ and we can always make this region small enough that

$$z\lambda^2 + (1-z)^2 m^2 > xyp^2 \implies \Delta > 0. \quad (190)$$

I write

$$\mathbb{I} \sim -\frac{i\pi^2}{2\Delta}, \quad (191)$$

where $\sim$ in this section means: exact in the IR divergent region.

$$T_1 \sim -\frac{i\pi^2}{2} \vec{A} \cdot \int_0^1 dx dy dz \delta(x+y+z-1) \vec{K} \frac{\sigma}{\Delta} \quad (192)$$

$$T_1 \sim \frac{\pi^2}{2i} \vec{A} \cdot \int_0^1 dx dy \int_0^1 dz \delta(x+y+z-1) \frac{[(1-2y)\vec{p} + 2z\vec{P}](1+z)^2 m^2 - (1+y+z-y^2-yz)p^2}{z\lambda^2 + (1-z)^2 m^2 - xyp^2} \quad (193)$$

$$\int_0^1 dx dy dz f(x, y, z) \delta(x+y+z-1) = \int_0^1 dx \int_0^{1-x} dy f(x, y, 1-x-y) \quad (194)$$

Here and in Sec. 3.3.6 I am only interested in the region near $x = y = 0$ so I need the simpler identity

$$\int_0^\delta dx dy \int_0^1 dz f(x, y, z) \delta(x+y+z-1) = \int_0^\delta dx dy f(x, y, 1-x-y), \quad \delta \rightarrow 0_+. \quad (195)$$

$$T_1 \sim \frac{\pi^2}{2i} \vec{A} \cdot \int_0^\delta dx dy [(1-2y)\vec{p} + 2(1-x-y)\vec{P}] \frac{(2-x-y)^2 m^2 - (2-x-y+xy)p^2}{(1-x-y)\lambda^2 + (x+y)^2 m^2 - xyp^2} \quad (196)$$

$$x = X\delta, \quad y = Y\delta \quad (197)$$

$$T_1 \sim \frac{\pi^2 \delta^2}{2i} \vec{A} \cdot \int_0^1 dX dY [(1-2\delta Y)\vec{p} + 2(1-\delta X-\delta Y)\vec{P}] \frac{(2-\delta X-\delta Y)^2 m^2 - (2-\delta X-\delta Y+\delta^2 XY)p^2}{(1-\delta X-\delta Y)\lambda^2 + \delta^2(X+Y)^2 m^2 - \delta^2 XY p^2} \quad (198)$$

$$T_1 \sim -i\pi^2 \delta^2 \vec{A} \cdot (\vec{p} + 2\vec{P})(2m^2 - p^2) \int_0^1 \frac{dX dY}{\lambda^2 + \delta^2(X+Y)^2 m^2 - \delta^2 XY p^2} \quad (199)$$

$$T_1 \sim i\pi^2 \vec{A} \cdot (\vec{p} + 2\vec{P})(p^2 - 2m^2) \int_0^1 \frac{dX dY}{(\lambda/\delta)^2 + (X+Y)^2 m^2 - XY p^2} \quad (200)$$

$$c = \frac{p^2}{m^2}, \quad L = \frac{\lambda}{m\delta}, \quad \mathbb{I}_1 = \int_0^1 \frac{dX dY}{L^2 + (X+Y)^2 - cXY} \quad (201)$$

$$T_1 \sim i\pi^2 \vec{A} \cdot (\vec{p} + 2\vec{P})(c-2)\mathbb{I}_1 \quad (202)$$

IR problems are difficult because there are two limits, $\delta, \lambda \rightarrow 0_+$, and their order matters. We isolated the IR divergence

inside $\mathbb{I}_1$. This divergence can only exist at $X = Y = L = 0$, which implies $L \rightarrow 0_+$: thus we take $\lambda \rightarrow 0_+$ first.

$$\mathbb{I}_1 \sim \int_0^{\pi/2} d\phi \int_0^1 dr \frac{r}{L^2 + (X+Y)^2 - cXY}, \quad X = r \cos \phi, \quad Y = r \sin \phi \quad (203)$$

I integrate over the quarter circle enclosed by the original domain: I can throw away parts of the domain as long as I keep the IR divergence, which is the only thing I track in detail.

$$\mathbb{I}_1 \sim \int_0^{\pi/2} d\phi \int_0^1 dr \frac{r}{L^2 + r^2(\cos \phi + \sin \phi)^2 - cr^2 \cos \phi \sin \phi} \quad (204)$$

$$\mathbb{I}_1 \sim \int_0^{\pi/2} d\phi \int_0^1 dr \frac{r}{L^2 + f(\phi)r^2} \quad (205)$$

$$f(\phi) = 1 + (2-c) \sin \phi \cos \phi \quad (206)$$

$$R(r) = \int dr \frac{r}{L^2 + f(\phi)r^2} = \frac{\ln[L^2 + f(\phi)r^2]}{2f(\phi)} \quad (207)$$

$$\mathbb{I}_1 \sim \int_0^{\pi/2} d\phi [R(1) - R(0)] \quad (208)$$

$$\mathbb{I}_1 \sim \int_0^{\pi/2} d\phi R(1) - \int_0^{\pi/2} d\phi R(0) \quad (209)$$

$$\mathbb{I}_1 \sim \frac{1}{2} \int_0^{\pi/2} d\phi \frac{\ln[L^2 + f(\phi)]}{f(\phi)} - \frac{\ln(L^2)}{2} \int_0^{\pi/2} \frac{d\phi}{f(\phi)} \quad (210)$$

$$\Phi(c) = \int_0^{\pi/2} \frac{d\phi}{1 + (2-c) \sin \phi \cos \phi} \quad (211)$$

$$\mathbb{I}_1 \sim \frac{1}{2} \int_0^{\pi/2} d\phi \frac{\ln[L^2 + f(\phi)]}{f(\phi)} - \Phi(c) \ln L \quad (212)$$

I make the limit $\lambda \rightarrow 0_+$ explicit.

$$\mathbb{I}_1 \sim \frac{1}{2} \lim_{\lambda \rightarrow 0_+} \int_0^{\pi/2} d\phi \frac{\ln[L^2 + f(\phi)]}{f(\phi)} - \Phi(c) \lim_{\lambda \rightarrow 0_+} \ln L \quad (213)$$

$$\mathbb{I}_1 \sim \frac{1}{2} \int_0^{\pi/2} d\phi \frac{\ln[f(\phi)]}{f(\phi)} - \Phi(c) \lim_{\lambda \rightarrow 0_+} \ln L \quad (214)$$

I ignore the IR finite term.

$$\mathbb{I}_1 \sim -\Phi(c) \lim_{\lambda \rightarrow 0_+} \ln L = -\Phi(c) \lim_{\lambda \rightarrow 0_+} \ln \left(\frac{\lambda}{m\delta} \right) \quad (215)$$

$$\mathbb{I}_1 \sim -\Phi(c) \lim_{\lambda \rightarrow 0_+} \ln \left(\frac{\lambda}{m} \right) + \Phi(c) \ln \delta \quad (216)$$

The $\ln \delta$ term diverges as $\delta \rightarrow 0_+$. But δ terms are created in pairs by splitting integrals, $\int_0^\infty = \int_0^\delta + \int_\delta^\infty$, so they will cancel out in pairs.

$$\mathbb{I}_1 \sim -\Phi(c) \lim_{\lambda \rightarrow 0_+} \ln \left(\frac{\lambda}{m} \right) \quad (217)$$

$$T_1 \sim -i\pi^2 \Phi(c) \vec{A} \cdot (\vec{p} + 2\vec{P})(c-2) \lim_{\lambda \rightarrow 0_+} \ln \left(\frac{\lambda}{m} \right) \quad (218)$$

I restore the IR finite pieces and make the limit $\lambda \rightarrow 0_+$ implicit.

$$T_1 = -i\pi^2 \Phi(c) \vec{A} \cdot (\vec{p} + 2\vec{P})(c-2) \ln \left(\frac{\lambda}{m} \right) + \mathcal{U} \quad (219)$$

3.3.8. Final

$$\boxed{\mathcal{M}_v = \frac{e^3}{8\pi^2} \Phi(c) \vec{A} \cdot (\vec{p} + 2\vec{P})(c-2) \ln \left(\frac{\lambda}{m} \right) + e^3 \mathcal{U}} \quad (220)$$

3.4. Final

$$\mathcal{M} = \mathcal{M}_0 + \mathcal{M}_2 + \mathcal{M}_v \quad (221)$$

$$\boxed{\mathcal{M}_0 = -e(\vec{p} + 2\vec{P}) \cdot \vec{A}} \quad (222)$$

$$\mathcal{M}_2 = e^3 \mathcal{U} \quad (223)$$

$$\mathcal{M}_v = \frac{e^3}{8\pi^2} \Phi(c) \vec{A} \cdot (\vec{p} + 2\vec{P})(c-2) \ln \left(\frac{\lambda}{m} \right) + e^3 \mathcal{U} \quad (224)$$

$$\boxed{\mathcal{M} = \mathcal{M}_0 + \frac{e^3}{8\pi^2} \Phi(c) \vec{A} \cdot (\vec{p} + 2\vec{P})(c-2) \ln \left(\frac{\lambda}{m} \right) + e^3 \mathcal{U}} \quad (225)$$

3.5. Simplify

Eq. 225 is exact, but too complicated for a homework. We will simplify it. We ignore the IR finite parts, since they

will not affect the IR divergent part of the cross section to the order of interest.

$$\mathcal{M} = \mathcal{M}_0 + \frac{e^3}{8\pi^2} \Phi(c) \vec{A} \cdot (\vec{p} + 2\vec{P})(c-2) \ln \left(\frac{\lambda}{m} \right) \quad (226)$$

Also I assume the 4-momentum transferred to the scalar particle is spacelike, $p^2 < 0$. Then the assumption in Eq. 190 is trivially satisfied. Spacelike momentum transfer would happen, for example, if the external electromagnetic field $\vec{A}$ was purely magnetic: then it would only change the particle's direction, not its speed. With $p^2 < 0 \implies c < 0$ we can integrate $\Phi(c)$ analytically.

$$\Phi(c) = \frac{1}{\sqrt{(c-4)c}} \ln \left[\frac{c-2-\sqrt{(c-4)c}}{c-2+\sqrt{(c-4)c}} \right] \quad (227)$$

$$\mathcal{M} = \mathcal{M}_0 + \frac{e^3}{8\pi^2} \frac{\vec{A} \cdot (\vec{p} + 2\vec{P})}{\sqrt{(c-4)c}} \ln \left[\frac{c-2-\sqrt{(c-4)c}}{c-2+\sqrt{(c-4)c}} \right] (c-2) \ln \left(\frac{\lambda}{m} \right) \quad (228)$$

I assume the momentum transfer is large: $|c| \gg 1$.

$$\lim_{c \rightarrow -\infty} \ln \left[\frac{c-2-\sqrt{(c-4)c}}{c-2+\sqrt{(c-4)c}} \right] = \ln c^2 = 2 \ln |c| = 2 \ln(-c) \quad (229)$$

$$\mathcal{M} = \mathcal{M}_0 + \frac{e^3}{8\pi^2} \frac{\vec{A} \cdot (\vec{p} + 2\vec{P})}{\sqrt{c^2}} 2c \ln(-c) \ln \left(\frac{\lambda}{m} \right) \quad (230)$$

$$\mathcal{M} = \mathcal{M}_0 + \frac{e^3}{4\pi^2} \frac{c}{|c|} \vec{A} \cdot (\vec{p} + 2\vec{P}) \ln(-c) \ln \left(\frac{\lambda}{m} \right) \quad (231)$$

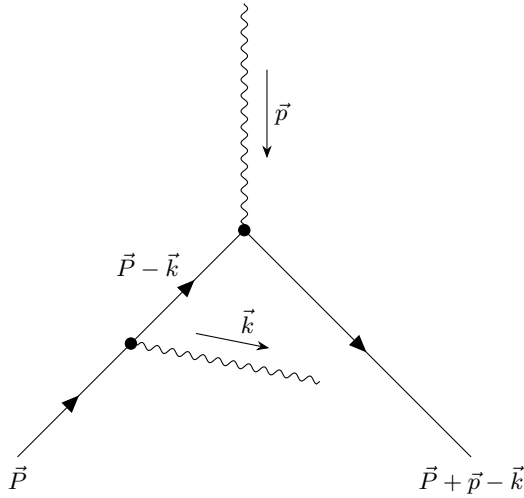
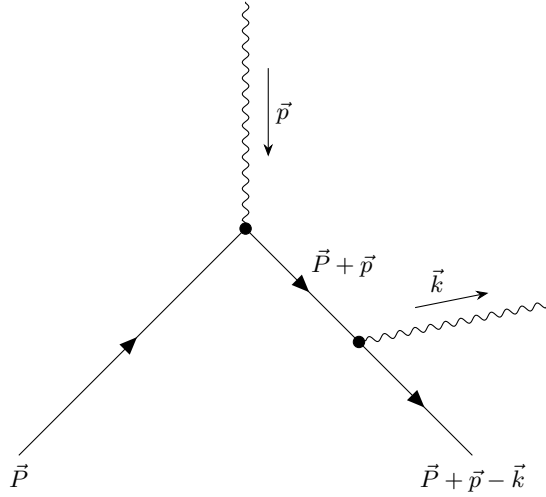
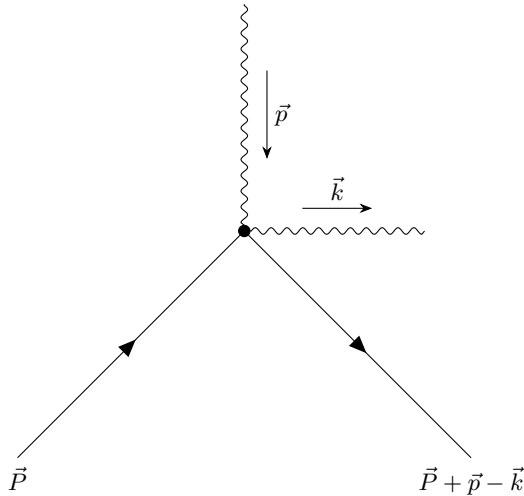
$$\frac{c}{|c|} = -1 \quad (232)$$

$$\mathcal{M} = \mathcal{M}_0 - \frac{e^3}{4\pi^2} \vec{A} \cdot (\vec{p} + 2\vec{P}) \ln(-c) \ln \left(\frac{\lambda}{m} \right) \quad (233)$$

$$\mathcal{M} = \mathcal{M}_0 - \frac{e^3}{4\pi^2} \vec{A} \cdot (\vec{p} + 2\vec{P}) \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \quad (234)$$

$$\mathcal{M}_0 = -e(\vec{p} + 2\vec{P}) \cdot \vec{A} \quad (235)$$

$$\boxed{\frac{\mathcal{M}}{\mathcal{M}_0} = 1 + \frac{e^2}{4\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right)} \quad (236)$$

FIG. 10. $\mathfrak{V}[1]$ FIG. 11. $\mathfrak{V}[2]$ FIG. 12. $\mathfrak{V}[3]$

4. BREMSSTRAHLUNG S-MATRIX

$$i\mathcal{M}_B = \mathfrak{V}[1] + \mathfrak{V}[2] + \mathfrak{V}[3] \quad (237)$$

The diagrams are in Figs. 10-12.

$$\mathfrak{V}[1] = ie[-\vec{P} - (\vec{P} - \vec{k})] \cdot \vec{\epsilon}^*(\vec{k}) \frac{i}{(\vec{P} - \vec{k})^2 - m^2 + i\epsilon} ie[-(\vec{P} - \vec{k}) - (\vec{P} + \vec{p} - \vec{k})] \cdot \vec{A} \quad (238)$$

$$\mathfrak{V}[1] = -ie^2 \frac{(2\vec{k} - \vec{p} - 2\vec{P}) \cdot \vec{A}}{(\vec{P} - \vec{k})^2 - m^2} (\vec{k} - 2\vec{P}) \cdot \vec{\epsilon}^*(\vec{k}) \quad (239)$$

$$\mathfrak{V}[1] = -ie^2 \frac{(2\vec{k} - \vec{p} - 2\vec{P}) \cdot \vec{A}}{P^2 + k^2 - 2\vec{P} \cdot \vec{k} - m^2} (\vec{k} - 2\vec{P}) \cdot \vec{\epsilon}^*(\vec{k}) \quad (240)$$

We can't forget the fictitious photon mass λ .

$$P^2 = m^2, \quad k^2 = \lambda^2 \quad (241)$$

$$\mathfrak{V}[1] = ie^2 \frac{(2\vec{k} - \vec{p} - 2\vec{P}) \cdot \vec{A}}{\lambda^2 - 2\vec{P} \cdot \vec{k}} (2\vec{P} - \vec{k}) \cdot \vec{\epsilon}^*(\vec{k}) \quad (242)$$

$$\mathfrak{V}[2] = ie[-\vec{P} - (\vec{p} + \vec{P})] \cdot \vec{A} \frac{i}{(\vec{p} + \vec{P})^2 - m^2 + i\epsilon} ie[-(\vec{p} + \vec{P}) - (\vec{P} + \vec{p} - \vec{k})] \cdot \vec{\epsilon}^*(\vec{k}) \quad (243)$$

$$\mathfrak{V}[2] = ie^2 \frac{(\vec{p} + 2\vec{P}) \cdot \vec{A}}{(\vec{p} + \vec{P})^2 - m^2} (\vec{k} - 2\vec{p} - 2\vec{P}) \cdot \vec{\epsilon}^*(\vec{k}) \quad (244)$$

$$\mathfrak{V}[2] = ie^2 \frac{(\vec{p} + 2\vec{P}) \cdot \vec{A}}{p^2 + 2\vec{p} \cdot \vec{P}} (\vec{k} - 2\vec{p} - 2\vec{P}) \cdot \vec{\epsilon}^*(\vec{k}) \quad (245)$$

The on-shell momenta satisfy

$$P^2 = (\vec{P} + \vec{p} - \vec{k})^2 = m^2 \implies 2\vec{P} \cdot \vec{p} = 2\vec{k} \cdot (\vec{p} + \vec{P}) - p^2 - \lambda^2. \quad (246)$$

$$\mathfrak{V}[2] = ie^2 \frac{(\vec{p} + 2\vec{P}) \cdot \vec{A}}{2\vec{k} \cdot (\vec{p} + \vec{P}) - \lambda^2} (\vec{k} - 2\vec{p} - 2\vec{P}) \cdot \vec{\epsilon}^*(\vec{k}) \quad (247)$$

$$\mathfrak{V}[3] = 2ie^2 \eta^{\mu\nu} \epsilon_\mu^*(\vec{k}) A_\nu \quad (248)$$

$$\mathfrak{V}[3] = 2ie^2 \vec{\epsilon}^*(\vec{k}) \cdot \vec{A} \quad (249)$$

$$\mathcal{M}_B = e^2 \left[\frac{(2\vec{k} - \vec{p} - 2\vec{P}) \cdot \vec{A}}{\lambda^2 - 2\vec{P} \cdot \vec{k}} (2\vec{P} - \vec{k}) + 2\vec{A} + \frac{(\vec{p} + 2\vec{P}) \cdot \vec{A}}{2\vec{k} \cdot (\vec{p} + \vec{P}) - \lambda^2} (\vec{k} - 2\vec{p} - 2\vec{P}) \right] \cdot \vec{\epsilon}^*(\vec{k}) \quad (250)$$

4.1. Soft Photon Limit

The emitted photons are soft (low energy-momentum) so I can take $\vec{k} \rightarrow \vec{0} \implies \lambda \rightarrow 0$ in Eq. 250, as long as I don't change the IR divergence structure.

$$\mathcal{M}_B = 2e^2 \vec{\epsilon}^*(\vec{k}) \cdot \left[\left(\frac{\vec{P}}{2\vec{P} \cdot \vec{k} - \lambda^2} - \frac{\vec{p} + \vec{P}}{2\vec{k} \cdot (\vec{p} + \vec{P}) - \lambda^2} \right) (\vec{p} + 2\vec{P}) \cdot \vec{A} + \vec{A} \right] \quad (251)$$

$$\vec{k} = (E, \mathbf{k}), \quad E = \sqrt{|\mathbf{k}|^2 + \lambda^2} \quad (252)$$

I assume $|\mathbf{k}| \sim \lambda$: the two are similar in magnitude. This implies $|\vec{k} \cdot \vec{P}| \gg \lambda^2$.

$$\mathcal{M}_B = e^2 \left[\left(\frac{\vec{P}}{\vec{P} \cdot \vec{k}} - \frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})} \right) (\vec{p} + 2\vec{P}) \cdot \vec{A} + 2\vec{A} \right] \cdot \vec{\epsilon}^*(\vec{k}) \quad (253)$$

$$\mathcal{M}_0 = -e(\vec{p} + 2\vec{P}) \cdot \vec{A} \quad (254)$$

$$\mathcal{M}_B = e \left[2e\vec{A} - \mathcal{M}_0 \left(\frac{\vec{P}}{\vec{P} \cdot \vec{k}} - \frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})} \right) \right] \cdot \vec{\epsilon}^*(\vec{k}) \quad (255)$$

5. KINEMATICS

I set up the kinematics to simplify the cross-sections. The initial momentum is

$$\vec{P} = (E_P, \mathbf{P}) = (E_P, 0, 0, P_z), \quad E_P = \sqrt{P_z^2 + m^2}, \quad P_z > 0. \quad (256)$$

The momentum transferred with the field is

$$\vec{p} = (0, \mathbf{p}). \quad (257)$$

Since I assumed $\vec{p}$ is spacelike I can, without loss of generality, choose a frame where $p^0 = 0$. The final scalar momentum is

$$\vec{K} = (E_K, \mathbf{K}), \quad E_K = \sqrt{|\mathbf{K}|^2 + m^2}. \quad (258)$$

The emitted soft photon momentum is

$$\vec{k} = (E, \mathbf{k}), \quad E = \sqrt{|\mathbf{k}|^2 + \lambda^2}. \quad (259)$$

6. LOWEST ORDER DECAY RATE

$$\mathcal{M}_0 = -e(\vec{p} + 2\vec{P}) \cdot \vec{A} \quad (260)$$

$\vec{A}$ is actually the Fourier transform of the electromagnetic field; it can be complex.

$$\mathcal{M}_0^* = -e(\vec{p} + 2\vec{P}) \cdot \vec{A}^* \quad (261)$$

$$|\mathcal{M}_0|^2 = e^2(\vec{p} + 2\vec{P}) \cdot \vec{A}(\vec{p} + 2\vec{P}) \cdot \vec{A}^* \quad (262)$$

From Fig. 4 $\vec{K} = \vec{p} + \vec{P} \implies \vec{p} = \vec{K} - \vec{P}$.

$$|\mathcal{M}_0|^2 = e^2(\vec{K} + \vec{P}) \cdot \vec{A}(\vec{K} + \vec{P}) \cdot \vec{A}^* = e^2|(\vec{K} + \vec{P}) \cdot \vec{A}|^2 \quad (263)$$

$\mathcal{M}_0$ describes a 1 to 1 scattering process; the momentum exchange with the external field is not a particle. I cannot use Schwartz's 2 to n cross section equation. I can use his decay rate (Eq. 5.24). The decay rate and cross section are proportional. I will demonstrate the IR divergence cancellation with decay rates instead of cross sections.

$$d\Gamma_0 = \frac{|\mathcal{M}_0|^2}{2E_P} \frac{d^3K}{(2\pi)^3} \frac{(2\pi)^4}{2E_K} \delta^{(4)}(\vec{P} + \vec{p} - \vec{K}) \quad (264)$$

$$d\Gamma_0 = \frac{\pi |\mathcal{M}_0|^2}{2E_P E_K} d^3K \delta^{(4)}(\vec{P} + \vec{p} - \vec{K}) \quad (265)$$

$$\Gamma_0 = \frac{\pi}{2E_P} \int d^3K \frac{|\mathcal{M}_0|^2}{E_K} \delta^{(4)}(\vec{P} + \vec{p} - \vec{K}) \quad (266)$$

$$\Gamma_0 = \frac{\pi}{2E_P} \int d^3K \frac{|\mathcal{M}_0|^2}{E_K} \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{K}) \delta(E_P + p^0 - E_K) \quad (267)$$

$$p^0 = 0 \quad (268)$$

$$\Gamma_0 = \frac{\pi}{2E_P} \int d^3K \frac{|\mathcal{M}_0|^2}{E_K} \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{K}) \delta(E_P - E_K) \quad (269)$$

$$\Gamma_0 = \frac{\pi e^2}{2E_P} \int d^3K \frac{|(\vec{K} + \vec{P}) \cdot \vec{A}|^2}{\sqrt{|\mathbf{K}|^2 + m^2}} \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{K}) \delta\left(E_P - \sqrt{|\mathbf{K}|^2 + m^2}\right) \quad (270)$$

$$\Gamma_0 = \frac{\pi e^2}{2E_P} \int d^3K \frac{|[(E_K, \mathbf{K}) + \vec{P}] \cdot \vec{A}|^2}{\sqrt{|\mathbf{K}|^2 + m^2}} \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{K}) \delta\left(E_P - \sqrt{|\mathbf{K}|^2 + m^2}\right) \quad (271)$$

$$\Gamma_0 = \frac{\pi e^2}{2E_P} \int d^3K \frac{|[(\sqrt{|\mathbf{K}|^2 + m^2}, \mathbf{K}) + \vec{P}] \cdot \vec{A}|^2}{\sqrt{|\mathbf{K}|^2 + m^2}} \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{K}) \delta\left(E_P - \sqrt{|\mathbf{K}|^2 + m^2}\right) \quad (272)$$

$$\Gamma_0 = \frac{\pi e^2}{2E_P} \frac{[(\sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2}, \mathbf{P} + \mathbf{p}) + \vec{P}] \cdot \vec{A}]^2}{\sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2} \delta(E_P - \sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2})} \quad (273)$$

$$\mathcal{E} = \sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2} \quad (274)$$

$$\Gamma_0 = \frac{\pi e^2}{2E_P \mathcal{E}} [(\mathcal{E}, \mathbf{P} + \mathbf{p}) + \vec{P}] \cdot \vec{A}]^2 \delta(E_P - \mathcal{E}) \quad (275)$$

$$\Gamma_0 = \frac{\pi e^2}{2E_P \mathcal{E}} |(\vec{P} + \vec{p} + \vec{P}) \cdot \vec{A}|^2 \delta(E_P - \mathcal{E}) \quad (276)$$

$$\Gamma_0 = \frac{\pi e^2}{2E_P \mathcal{E}} |(\vec{p} + 2\vec{P}) \cdot \vec{A}|^2 \delta(E_P - \mathcal{E}) \quad (277)$$

$$|\mathcal{M}_0|^2 = e^2 |(\vec{p} + 2\vec{P}) \cdot \vec{A}|^2 \quad (278)$$

$$\boxed{\Gamma_0 = \frac{\pi |\mathcal{M}_0|^2}{2E_P \mathcal{E}} \delta(E_P - \mathcal{E})} \quad (279)$$

7. $1 \rightarrow 1$ DECAY RATE

$$\mathcal{M} = \mathcal{M}_0 \left[1 + \frac{e^2}{4\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \right] \quad (280)$$

$$|\mathcal{M}|^2 = |\mathcal{M}_0|^2 \left[1 + \frac{e^2}{4\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \right]^2 \quad (281)$$

$$|\mathcal{M}|^2 = |\mathcal{M}_0|^2 \left[1 + \frac{e^4}{16\pi^4} \ln \left(-\frac{p^2}{m^2} \right)^2 \ln \left(\frac{\lambda}{m} \right)^2 + \frac{e^2}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \right] \quad (282)$$

I only want the leading order IR divergence and drop all other terms.

$$|\mathcal{M}|^2 = |\mathcal{M}_0|^2 \frac{e^2}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \quad (283)$$

$$\Gamma_0 = \frac{\pi}{2E_P} \int d^3K \frac{|\mathcal{M}_0|^2}{E_K} \delta^{(4)}(\vec{P} + \vec{p} - \vec{K}) \quad (284)$$

$$\Gamma = \frac{\pi}{2E_P} \int d^3K \frac{|\mathcal{M}|^2}{E_K} \delta^{(4)}(\vec{P} + \vec{p} - \vec{K}) \quad (285)$$

$$\Gamma = \frac{e^2}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \frac{\pi}{2E_P} \int d^3K \frac{|\mathcal{M}_0|^2}{E_K} \delta^{(4)}(\vec{P} + \vec{p} - \vec{K}) \quad (286)$$

$$\boxed{\Gamma = \frac{e^2}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \Gamma_0} \quad (287)$$

8. BREMSSTRAHLUNG DECAY RATE

8.1. $|\mathcal{M}_B|^2$

$$\vec{J} = \frac{\vec{P}}{\vec{P} \cdot \vec{k}} - \frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})} \quad (288)$$

$$\mathcal{M}_B = e(2e\vec{A} - \mathcal{M}_0\vec{J}) \cdot \vec{\epsilon}^*(\vec{k}) = e(2eA^\alpha - \mathcal{M}_0J^\alpha) \epsilon_\alpha^*(\vec{k}) \quad (289)$$

$$\mathcal{M}_B^* = e(2eA^{\beta*} - \mathcal{M}_0^*J^\beta) \epsilon_\beta(\vec{k}) \quad (290)$$

$$|\mathcal{M}_B|^2 = \mathcal{M}_B \mathcal{M}_B^* \quad (291)$$

$$|\mathcal{M}_B|^2 = e^2(2eA^\alpha - \mathcal{M}_0J^\alpha)(2eA^{\beta*} - \mathcal{M}_0^*J^\beta) \epsilon_\alpha^* \epsilon_\beta \quad (292)$$

$$|\mathcal{M}_B|^2 = e^2[4e^2A^\alpha A^{\beta*} - 2e(A^\alpha \mathcal{M}_0^*J^\beta + A^{\beta*} \mathcal{M}_0J^\alpha) + |\mathcal{M}_0|^2 J^\alpha J^\beta] \epsilon_\alpha^* \epsilon_\beta \quad (293)$$

I drop the IR finite term.

$$|\mathcal{M}_B|^2 = e^2$$

$$[|\mathcal{M}_0|^2 J^\alpha J^\beta - 2e(A^\alpha \mathcal{M}_0^*J^\beta + A^{\beta*} \mathcal{M}_0J^\alpha)] \epsilon_\alpha^* \epsilon_\beta \quad (294)$$

8.2. Polarization

$$(|\mathcal{M}_B|^s)^2 = e^2$$

$$[|\mathcal{M}_0|^2 J^\alpha J^\beta - 2e(A^\alpha \mathcal{M}_0^*J^\beta + A^{\beta*} \mathcal{M}_0J^\alpha)] \epsilon_\alpha^{s*} \epsilon_\beta^s \quad (295)$$

Here s is the polarization index, which I have suppressed up to now. I sum over final polarizations

$$\mathcal{A} = \sum_{s=1}^2 (|\mathcal{M}_B|^s)^2. \quad (296)$$

$$\mathcal{A} = e^2 [|\mathcal{M}_0|^2 J^\alpha J^\beta - 2e(A^\alpha \mathcal{M}_0^*J^\beta + A^{\beta*} \mathcal{M}_0J^\alpha)] \sum_{s=1}^2 \epsilon_\alpha^{s*} \epsilon_\beta^s \quad (297)$$

From Schwartz Eq. (13.110)

$$\sum_{s=1}^2 \epsilon_\mu^{s*}(\vec{p}) \epsilon_\nu^s(\vec{p}) = \frac{p_\mu \bar{p}_\nu + \bar{p}_\mu p_\nu}{2E^2} - \eta_{\mu\nu}, \quad p^\mu = (E, \mathbf{p}), \quad \bar{p}^\mu = (E, -\mathbf{p}). \quad (298)$$

Because $\mathcal{M}_B$ satisfies the Ward identity, this simplifies to

$$\sum_{s=1}^2 \epsilon_\mu^{s*}(\vec{p}) \epsilon_\nu^s(\vec{p}) \rightarrow -\eta_{\mu\nu}. \quad (299)$$

My expression for $|\mathcal{M}_B|^2$ (Eq. 294) appears to violate the Ward identity, but that's only because I have ignored IR finite terms.

$$\mathcal{A} = e^2 [2e(A^\alpha \mathcal{M}_0^* J^\beta + A^{\beta*} \mathcal{M}_0 J^\alpha) - |\mathcal{M}_0|^2 J^\alpha J^\beta] \eta_{\alpha\beta} \quad (300)$$

$$\mathcal{A} = e^2 [2e(\mathcal{M}_0^* \vec{A} \cdot \vec{J} + \mathcal{M}_0 \vec{A}^* \cdot \vec{J}) - |\mathcal{M}_0|^2 \vec{J} \cdot \vec{J}] \quad (301)$$

$$\boxed{\mathcal{A} = e^2 [2e(\mathcal{M}_0^* \vec{A} \cdot \vec{J} + \mathcal{M}_0 \vec{A}^* \cdot \vec{J}) - |\mathcal{M}_0|^2 J^2]} \quad (302)$$

8.3. Grouping Decay Rate Terms

$$d\Gamma_B = \frac{\mathcal{A}}{2E_P} \frac{d^3 K d^3 k}{(2\pi)^6} \frac{1}{4E_K E} (2\pi)^4 \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \quad (303)$$

$$d\Gamma_B = \frac{\mathcal{A} d^3 K d^3 k}{32\pi^2 E_P E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \quad (304)$$

$$\Gamma_B = \int d^3 K d^3 k \mathcal{A} \frac{\delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K})}{32\pi^2 E_P E_K E} \quad (305)$$

$$\mathcal{A} = e^2 [2e(\mathcal{M}_0^* \vec{A} \cdot \vec{J} + \mathcal{M}_0 \vec{A}^* \cdot \vec{J}) - |\mathcal{M}_0|^2 J^2] \quad (306)$$

$$\begin{aligned} 32\pi^2 \frac{\Gamma_B}{e^2} &= 2e \left[\int d^3 K d^3 k \frac{\mathcal{M}_0^* \vec{A} \cdot \vec{J}}{E_P E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \right. \\ &\quad \left. + \int d^3 K d^3 k \frac{\mathcal{M}_0 \vec{A}^* \cdot \vec{J}}{E_P E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \right] \\ &\quad - \int d^3 K d^3 k \frac{|\mathcal{M}_0|^2 J^2}{E_P E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \end{aligned} \quad (307)$$

$$\mathcal{M}_0 = -e(\vec{p} + 2\vec{P}) \cdot \vec{A} \quad (308)$$

$\mathcal{M}_0$ does not depend on $\vec{K}$ so I take it outside the integrals. I can write it in terms of $\vec{K}$, like I did in Sec. 6, but that will only make the algebra longer.

$$\begin{aligned} 32\pi^2 \frac{\Gamma_B}{e^2} &= 2e \left[\mathcal{M}_0^* \int d^3 K d^3 k \frac{\vec{A} \cdot \vec{J}}{E_P E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \right. \\ &\quad \left. + \mathcal{M}_0 \int d^3 K d^3 k \frac{\vec{A}^* \cdot \vec{J}}{E_P E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \right] \\ &\quad - |\mathcal{M}_0|^2 \int d^3 K d^3 k \frac{\delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) J^2}{E_P E_K E} \end{aligned} \quad (309)$$

$$E = \sqrt{|\mathbf{k}|^2 + \lambda^2}, \quad E_K = \sqrt{|\mathbf{K}|^2 + m^2}, \quad E_P = \sqrt{P_z^2 + m^2} \quad (310)$$

$$\begin{aligned} 32\pi^2 E_P \frac{\Gamma_B}{e^2} &= 2e \left[\mathcal{M}_0^* \int d^3 K d^3 k \frac{\vec{A} \cdot \vec{J}}{E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \right. \\ &\quad \left. + \mathcal{M}_0 \int d^3 K d^3 k \frac{\vec{A}^* \cdot \vec{J}}{E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \right] \\ &\quad - |\mathcal{M}_0|^2 \int d^3 K d^3 k \frac{\delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) J^2}{E_K E} \end{aligned} \quad (311)$$

$$32\pi^2 E_P \frac{\Gamma_B}{e^2} = 2e(\mathcal{M}_0^* \mathcal{T} + \mathcal{M}_0 \mathcal{T}^*) - |\mathcal{M}_0|^2 \mathbb{T} \quad (312)$$

$$32\pi^2 E_P \frac{\Gamma_B}{e^2} = 4e \operatorname{Re}(\mathcal{M}_0^* \mathcal{T}) - |\mathcal{M}_0|^2 \mathbb{T} \quad (313)$$

$$\mathcal{T} = \int d^3 K d^3 k \frac{\vec{A} \cdot \vec{J}}{E_K E} \delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) \quad (314)$$

$$\mathbb{T} = \int d^3 K d^3 k \frac{\delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) J^2}{E_K E}$$

$$\mathbb{T} = \int d^3 K d^3 k \frac{\delta^{(4)}(\vec{P} + \vec{p} - \vec{k} - \vec{K}) J^2}{E \sqrt{|\mathbf{K}|^2 + m^2}} \quad (315)$$

$$\mathbb{T} = \int d^3 K d^3 k \frac{J^2 \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{k} - \mathbf{K})}{E \sqrt{|\mathbf{K}|^2 + m^2}} \delta(E_P + p^0 - E - E_K) \quad (316)$$

$$p^0 = 0 \quad (317)$$

$$\mathbb{T} = \int d^3 K d^3 k \frac{J^2 \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{k} - \mathbf{K})}{E \sqrt{|\mathbf{K}|^2 + m^2}} \delta(E_P - E - E_K) \quad (318)$$

$$\begin{aligned} \mathbb{T} &= \int d^3 K d^3 k \frac{J^2 \delta^{(3)}(\mathbf{P} + \mathbf{p} - \mathbf{k} - \mathbf{K})}{E \sqrt{|\mathbf{K}|^2 + m^2}} \\ &\quad \delta(E_P - E - \sqrt{|\mathbf{K}|^2 + m^2}) \end{aligned} \quad (319)$$

$$\mathbb{T} = \int d^3 k \frac{J^2 \delta(E_P - E - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \quad (320)$$

$$f(\mathbf{k}) = E_P - E - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2} \quad (321)$$

$$\mathbb{T} = \int d^3 k \frac{J^2 \delta[f(\mathbf{k})]}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \quad (322)$$

$$\begin{aligned} \mathcal{T} &= \int d^3 k \frac{\vec{A} \cdot \vec{J} \delta[f(\mathbf{k})]}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ \mathbb{T} &= \int d^3 k \frac{J^2 \delta[f(\mathbf{k})]}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \end{aligned} \quad (323)$$

8.4. IR Power Counting

$$32\pi^2 E_P \frac{\Gamma_B}{e^2} = 4e \operatorname{Re}(\mathcal{M}_0^* \mathcal{T}) - |\mathcal{M}_0|^2 \mathbb{T} \quad (324)$$

$$\begin{aligned} \mathcal{T} &= \int d^3k \frac{\vec{A} \cdot \vec{J} \delta[f(\mathbf{k})]}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ \mathbb{T} &= \int d^3k \frac{J^2 \delta[f(\mathbf{k})]}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \end{aligned} \quad (325)$$

$$E = \sqrt{|\mathbf{k}|^2 + \lambda^2}, \quad E_P = \sqrt{P_z^2 + m^2} \quad (326)$$

$$f(\mathbf{k}) = E_P - E - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2} \quad (327)$$

$$\vec{J} = \frac{\vec{P}}{\vec{P} \cdot \vec{k}} - \frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})} \quad (328)$$

The integrands in Eq. 325 diverge when $\vec{k} = \vec{0} \implies \lambda = 0$. This is the IR divergence. There could also be a divergence if $\vec{P} \cdot \vec{k} = 0$. In the rest frame of $\vec{P}$

$$\vec{P} \cdot \vec{k} = (m, \mathbf{0}) \cdot \vec{k} = mk^0 = m\sqrt{|\mathbf{k}|^2 + \lambda^2} > 0. \quad (329)$$

So $\vec{P} \cdot \vec{k} \neq 0$. The final scalar momentum (Fig. 10) is $\vec{p} + \vec{P} - \vec{k}$. Repeating the argument above yields

$$\vec{k} \cdot (\vec{p} + \vec{P} - \vec{k}) > 0. \quad (330)$$

This implies

$$\vec{k} \cdot (\vec{p} + \vec{P}) - k^2 = \vec{k} \cdot (\vec{p} + \vec{P}) - \lambda^2 > 0 \implies \vec{k} \cdot (\vec{p} + \vec{P}) > \lambda^2. \quad (331)$$

So $[\vec{k} \cdot (\vec{p} + \vec{P})]^{-1}$ can't diverge either. The only possible IR divergence is $\vec{k} \rightarrow \vec{0}$.

$$\begin{aligned} \mathcal{T} &= \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2 + \lambda^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{\sqrt{|\mathbf{k}|^2 + \lambda^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{\vec{A} \cdot \vec{P}}{\vec{P} \cdot \vec{k}} - \frac{\vec{A} \cdot (\vec{p} + \vec{P})}{\vec{k} \cdot (\vec{p} + \vec{P})} \right] \end{aligned} \quad (332)$$

When power counting I take $\vec{A} \cdot \vec{P} \rightarrow 1$ & $\vec{A} \cdot (\vec{p} + \vec{P}) \rightarrow 1$.

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2 + \lambda^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{\sqrt{|\mathbf{k}|^2 + \lambda^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{\vec{P} \cdot \vec{k}} - \frac{1}{\vec{k} \cdot (\vec{p} + \vec{P})} \right] \end{aligned} \quad (333)$$

$$\vec{P} = (E_P, \mathbf{P}), \quad \vec{p} = (0, \mathbf{p}), \quad \vec{k} = (E, \mathbf{k}) = (\sqrt{|\mathbf{k}|^2 + \lambda^2}, \mathbf{k}) \quad (334)$$

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2 + \lambda^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{\sqrt{|\mathbf{k}|^2 + \lambda^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{\vec{P} \cdot (\sqrt{|\mathbf{k}|^2 + \lambda^2}, \mathbf{k})} - \frac{1}{(\sqrt{|\mathbf{k}|^2 + \lambda^2}, \mathbf{k}) \cdot (\vec{p} + \vec{P})} \right] \end{aligned} \quad (335)$$

I set $\lambda \rightarrow 0$.

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{\sqrt{|\mathbf{k}|^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{\vec{P} \cdot (\sqrt{|\mathbf{k}|^2}, \mathbf{k})} - \frac{1}{(\sqrt{|\mathbf{k}|^2}, \mathbf{k}) \cdot (\vec{p} + \vec{P})} \right] \end{aligned} \quad (336)$$

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - |\mathbf{k}| - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}| \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{\vec{P} \cdot (|\mathbf{k}|, \mathbf{k})} - \frac{1}{(|\mathbf{k}|, \mathbf{k}) \cdot (\vec{p} + \vec{P})} \right] \end{aligned} \quad (337)$$

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - |\mathbf{k}| - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}| \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{(E_P, \mathbf{P}) \cdot (|\mathbf{k}|, \mathbf{k})} - \frac{1}{(|\mathbf{k}|, \mathbf{k}) \cdot [(0, \mathbf{p}) + (E_P, \mathbf{P})]} \right] \end{aligned} \quad (338)$$

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - |\mathbf{k}| - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}| \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{(E_P, \mathbf{P}) \cdot (|\mathbf{k}|, \mathbf{k})} - \frac{1}{(|\mathbf{k}|, \mathbf{k}) \cdot [(0, \mathbf{p}) + (E_P, \mathbf{P})]} \right] \end{aligned} \quad (339)$$

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - |\mathbf{k}| - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}| \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{(E_P, \mathbf{P}) \cdot (|\mathbf{k}|, \mathbf{k})} - \frac{1}{(|\mathbf{k}|, \mathbf{k}) \cdot (E_P, \mathbf{p} + \mathbf{P})} \right] \end{aligned} \quad (340)$$

$$\begin{aligned} \mathcal{T} &\sim \int d^3k \frac{\delta(E_P - |\mathbf{k}| - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}| \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ &\quad \left[\frac{1}{E_P |\mathbf{k}| - \mathbf{P} \cdot \mathbf{k}} - \frac{1}{E_P |\mathbf{k}| - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})} \right] \end{aligned} \quad (341)$$

$$\frac{1}{E_P |\mathbf{k}| - \mathbf{P} \cdot \mathbf{k}} - \frac{1}{E_P |\mathbf{k}| - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})} \rightarrow \frac{1}{|\mathbf{k}|} \quad (342)$$

$$\mathcal{T} \sim \int d^3k \frac{\delta(E_P - |\mathbf{k}| - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}|^2 \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \quad (343)$$

I take $\mathbf{k} \rightarrow \mathbf{0}$ because that is the region I want.

$$\mathcal{T} \sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2})}{|\mathbf{k}|^2 \sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2}} \quad (344)$$

$$\mathcal{T} \sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2})}{|\mathbf{k}|^2} \quad (345)$$

$$\mathcal{T} \sim \int d^3k \frac{\delta(C)}{|\mathbf{k}|^2} = \delta(C) \int \frac{d^3k}{|\mathbf{k}|^2} \sim \int \frac{d^3k}{|\mathbf{k}|^2} \sim \int_0^\infty dk \frac{k^2}{k^2} \quad (346)$$

C does not depend on $\mathbf{k}$.

$$\mathcal{T} \sim \int_0^\infty dk \sim \int_0^\delta dk = \delta, \quad \delta \rightarrow 0_+ \quad (347)$$

$\mathcal{T}$ is IR convergent.

$$\mathbb{T} = \int d^3k \frac{J^2 \delta[f(\mathbf{k})]}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \quad (348)$$

$$\mathbb{T} = \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2 + \lambda^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{\sqrt{|\mathbf{k}|^2 + \lambda^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} J^2 \quad (349)$$

$$\mathbb{T} \sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2 + \lambda^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{\sqrt{|\mathbf{k}|^2 + \lambda^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} J^2 \quad (350)$$

$$J^2 \sim \frac{1}{|\mathbf{k}|^2} \quad (351)$$

$$\mathbb{T} \sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2 + \lambda^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}|^2 \sqrt{|\mathbf{k}|^2 + \lambda^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \quad (352)$$

$$\mathbb{T} \sim \int d^3k \frac{\delta(E_P - |\mathbf{k}| - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{|\mathbf{k}|^3 \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \quad (353)$$

$$\mathbb{T} \sim \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2})}{|\mathbf{k}|^3 \sqrt{|\mathbf{P} + \mathbf{p}|^2 + m^2}} \sim \int d^3k \frac{\delta(C)}{|\mathbf{k}|^3} \quad (354)$$

$$\mathbb{T} \sim \delta(C) \int \frac{d^3k}{|\mathbf{k}|^3} \sim \int \frac{d^3k}{|\mathbf{k}|^3} \sim \int_0^\infty dk \frac{k^2}{k^3} = \int_0^\infty \frac{dk}{k} \quad (355)$$

$$\mathbb{T} \sim \int_\delta^1 \frac{dk}{k} = -\ln \delta, \quad \delta \rightarrow 0_+ \quad (356)$$

$\mathbb{T}$ has an IR divergence.

$$32\pi^2 E_P \frac{\Gamma_B}{e^2} = 4e \operatorname{Re}(\mathcal{M}_0^* \mathcal{T}) - |\mathcal{M}_0|^2 \mathbb{T} \quad (357)$$

I ignore IR convergent terms.

$$\boxed{\Gamma_B = -\frac{e^2 |\mathcal{M}_0|^2 \mathbb{T}}{32\pi^2 E_P}} \quad (358)$$

8.5. Isolate IR Divergence

$$\Gamma_B = -\frac{e^2 |\mathcal{M}_0|^2 \mathbb{T}}{32\pi^2 E_P} \quad (359)$$

$$\mathbb{T} = \int d^3k \frac{\delta[f(\mathbf{k})]}{E \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} J^2 \quad (360)$$

$$f(\mathbf{k}) = E_P - E - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2} \quad (361)$$

$$\vec{J} = \frac{\vec{P}}{\vec{P} \cdot \vec{k}} - \frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})}, \quad E = \sqrt{|\mathbf{k}|^2 + \lambda^2} \quad (362)$$

$$J^2 = \vec{J} \cdot \vec{J} \quad (363)$$

$$J^2 = \left[\frac{\vec{P}}{\vec{P} \cdot \vec{k}} - \frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})} \right]^2 \quad (364)$$

$$J^2 = \left(\frac{\vec{P}}{\vec{P} \cdot \vec{k}} \right)^2 + \left[\frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})} \right]^2 - \frac{2\vec{P}}{\vec{P} \cdot \vec{k}} \cdot \frac{\vec{p} + \vec{P}}{\vec{k} \cdot (\vec{p} + \vec{P})} \quad (365)$$

$$J^2 = \frac{P^2}{(\vec{P} \cdot \vec{k})^2} + \frac{(\vec{p} + \vec{P})^2}{[\vec{k} \cdot (\vec{p} + \vec{P})]^2} - \frac{2\vec{P} \cdot (\vec{p} + \vec{P})}{(\vec{P} \cdot \vec{k})[\vec{k} \cdot (\vec{p} + \vec{P})]} \quad (366)$$

$$J^2 = \frac{P^2}{(\vec{P} \cdot \vec{k})^2} + \frac{p^2 + P^2 + 2\vec{p} \cdot \vec{P}}{[\vec{k} \cdot (\vec{p} + \vec{P})]^2} - \frac{2\vec{P} \cdot \vec{p} + 2P^2}{(\vec{P} \cdot \vec{k})[\vec{k} \cdot (\vec{p} + \vec{P})]} \quad (367)$$

$$P^2 = (\vec{P} + \vec{p} - \vec{k})^2 = m^2 \implies 2\vec{P} \cdot \vec{p} = 2\vec{k} \cdot (\vec{p} + \vec{P}) - p^2 - \lambda^2. \quad (368)$$

$$J^2 = \frac{m^2}{(\vec{P} \cdot \vec{k})^2} + \frac{p^2 + m^2 + 2\vec{p} \cdot \vec{P}}{[\vec{k} \cdot (\vec{p} + \vec{P})]^2} - \frac{2\vec{P} \cdot \vec{p} + 2m^2}{(\vec{P} \cdot \vec{k})[\vec{k} \cdot (\vec{p} + \vec{P})]} \quad (369)$$

$$J^2 = \frac{m^2}{(\vec{P} \cdot \vec{k})^2} + \frac{p^2 + m^2 + 2\vec{k} \cdot (\vec{p} + \vec{P}) - p^2 - \lambda^2}{[\vec{k} \cdot (\vec{p} + \vec{P})]^2} - \frac{2\vec{k} \cdot (\vec{p} + \vec{P}) - p^2 - \lambda^2 + 2m^2}{(\vec{P} \cdot \vec{k})[\vec{k} \cdot (\vec{p} + \vec{P})]} \quad (370)$$

$$J^2 = \frac{m^2}{(\vec{P} \cdot \vec{k})^2} + \frac{m^2 - \lambda^2}{[\vec{k} \cdot (\vec{p} + \vec{P})]^2} + \frac{2}{\vec{k} \cdot (\vec{p} + \vec{P})} - \frac{2}{\vec{P} \cdot \vec{k}} + \frac{p^2 + \lambda^2 - 2m^2}{(\vec{P} \cdot \vec{k})[\vec{k} \cdot (\vec{p} + \vec{P})]} \quad (371)$$

I know from Sec. 8.4 that terms with only one power of $\vec{k}$ in the denominator are IR convergent. I ignore these.

$$J^2 = \frac{m^2}{(\vec{P} \cdot \vec{k})^2} + \frac{m^2 - \lambda^2}{[\vec{k} \cdot (\vec{p} + \vec{P})]^2} + \frac{p^2 + \lambda^2 - 2m^2}{(\vec{P} \cdot \vec{k})[\vec{k} \cdot (\vec{p} + \vec{P})]} \quad (372)$$

In the limit $\lambda^2 \ll m^2 \ll -p^2 \rightarrow \infty$

$$J^2 = \frac{m^2}{(\vec{P} \cdot \vec{k})^2} + \frac{m^2}{[\vec{k} \cdot (\vec{p} + \vec{P})]^2} + \frac{p^2}{(\vec{P} \cdot \vec{k})[\vec{k} \cdot (\vec{p} + \vec{P})]} \quad (373)$$

$$\vec{P} = (E_P, \mathbf{P}), \quad \vec{p} = (0, \mathbf{p}), \quad \vec{k} = (E, \mathbf{k}) \quad (374)$$

$$\vec{P} \cdot \vec{k} = (E_P, \mathbf{P}) \cdot (E, \mathbf{k}) = EE_P - \mathbf{k} \cdot \mathbf{P} \quad (375)$$

$$\vec{k} \cdot (\vec{p} + \vec{P}) = (E, \mathbf{k}) \cdot [(0, \mathbf{p}) + (E_P, \mathbf{P})] = (E, \mathbf{k}) \cdot (E_P, \mathbf{p} + \mathbf{P}) \\ = EE_P - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P}) \quad (376)$$

$$J^2 = \frac{m^2}{(EE_P - \mathbf{k} \cdot \mathbf{P})^2} + \frac{m^2}{[EE_P - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^2} \\ + \frac{p^2}{(EE_P - \mathbf{k} \cdot \mathbf{P})[EE_P - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]} \quad (377)$$

$$\mathbb{T} = \int d^3k \frac{\delta[f(\mathbf{k})]}{E\sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} J^2 \quad (378)$$

$$\mathbb{T} = \int d^3k \frac{\delta(E_P - E - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{E\sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ \left[\frac{m^2}{(EE_P - \mathbf{k} \cdot \mathbf{P})^2} + \frac{m^2}{[EE_P - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^2} \right. \\ \left. + \frac{p^2}{(EE_P - \mathbf{k} \cdot \mathbf{P})[EE_P - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (379)$$

$$\mathbb{T} = \int d^3k \frac{\delta(E_P - \sqrt{|\mathbf{k}|^2 + \lambda^2} - \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2})}{\sqrt{|\mathbf{k}|^2 + \lambda^2} \sqrt{|\mathbf{P} + \mathbf{p} - \mathbf{k}|^2 + m^2}} \\ \left[\frac{m^2}{(E_P \sqrt{|\mathbf{k}|^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})^2} \right. \\ + \frac{m^2}{[E_P \sqrt{|\mathbf{k}|^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^2} \\ \left. + \frac{p^2[E_P \sqrt{|\mathbf{k}|^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^{-1}}{(E_P \sqrt{|\mathbf{k}|^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})} \right] \quad (380)$$

$$\mathbb{T} = \int d^3k \frac{\delta[E_P - \sqrt{\mathbf{k}^2 + \lambda^2} - \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}]}{\sqrt{\mathbf{k}^2 + \lambda^2} \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}} \\ \left[\frac{m^2}{(E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})^2} \right. \\ + \frac{m^2}{[E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^2} \\ \left. + \frac{p^2[E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^{-1}}{(E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})} \right] \quad (381)$$

I introduce spherical coordinates.

$$\mathbf{k} = k\hat{\mathbf{r}}, \quad \hat{\mathbf{r}} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \quad (382)$$

$$\mathbb{T} = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\infty dk k^2 \\ \frac{\delta[E_P - \sqrt{\mathbf{k}^2 + \lambda^2} - \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}]}{\sqrt{\mathbf{k}^2 + \lambda^2} \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}} \\ \left[\frac{m^2}{(E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})^2} \right. \\ + \frac{m^2}{[E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^2} \\ \left. + \frac{p^2[E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^{-1}}{(E_P \sqrt{\mathbf{k}^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})} \right] \quad (383)$$

Don't confuse $k^2 = \mathbf{k}^2$ for $k^2 = \vec{k}^2$. Context will distinguish 3 from 4 vectors.

$$\mathbb{T} = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\infty dk k^2 \\ \frac{\delta[E_P - \sqrt{k^2 + \lambda^2} - \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}]}{\sqrt{k^2 + \lambda^2} \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}} \\ \left[\frac{m^2}{(E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})^2} \right. \\ + \frac{m^2}{[E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^2} \\ \left. + \frac{p^2[E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^{-1}}{(E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})} \right] \quad (384)$$

$$\mathbb{T} = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^{m\delta} dk k^2 \\ \frac{\delta[E_P - \sqrt{k^2 + \lambda^2} - \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}]}{\sqrt{k^2 + \lambda^2} \sqrt{(\mathbf{P} + \mathbf{p} - \mathbf{k})^2 + m^2}} \\ \left[\frac{m^2}{(E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})^2} \right. \\ + \frac{m^2}{[E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^2} \\ \left. + \frac{p^2[E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot (\mathbf{p} + \mathbf{P})]^{-1}}{(E_P \sqrt{k^2 + \lambda^2} - \mathbf{k} \cdot \mathbf{P})} \right], \quad \delta \rightarrow 0_+ \quad (385)$$

I ignore the IR convergent part ($m\delta < k < \infty$).

$$k = \lambda r, \quad \mathbf{k} = k\hat{\mathbf{r}} = \lambda r\hat{\mathbf{r}} = \lambda \mathbf{r}, \quad dk = \lambda dr, \quad r = \frac{k}{\lambda}, \\ r(0) = 0, \quad r(m\delta) = \frac{m}{\lambda} \delta = \rho \quad (386)$$

$$\mathbb{T} = \lambda^3 \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr r^2 \frac{\delta[E_P - \sqrt{\lambda^2 r^2 + \lambda^2} - \sqrt{(\mathbf{P} + \mathbf{p} - \lambda \mathbf{r})^2 + m^2}]}{\sqrt{\lambda^2 r^2 + \lambda^2} \sqrt{(\mathbf{P} + \mathbf{p} - \lambda \mathbf{r})^2 + m^2}} \left[\frac{m^2}{(E_P \sqrt{\lambda^2 r^2 + \lambda^2} - \lambda \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P \sqrt{\lambda^2 r^2 + \lambda^2} - \lambda \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2 [E_P \sqrt{\lambda^2 r^2 + \lambda^2} - \lambda \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^{-1}}{(E_P \sqrt{\lambda^2 r^2 + \lambda^2} - \lambda \mathbf{r} \cdot \mathbf{P})} \right] \quad (387)$$

$$\lambda > 0 \quad (388)$$

$$\mathbb{T} = \frac{\lambda^3}{\lambda^3} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr r^2 \frac{\delta[E_P - \lambda \sqrt{r^2 + 1} - \sqrt{(\mathbf{P} + \mathbf{p} - \lambda \mathbf{r})^2 + m^2}]}{\sqrt{r^2 + 1} \sqrt{(\mathbf{P} + \mathbf{p} - \lambda \mathbf{r})^2 + m^2}} \left[\frac{m^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (389)$$

$$\mathbb{T} = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr r^2 \frac{\delta[E_P - \lambda \sqrt{r^2 + 1} - \sqrt{(\mathbf{P} + \mathbf{p} - \lambda \mathbf{r})^2 + m^2}]}{\sqrt{r^2 + 1} \sqrt{(\mathbf{P} + \mathbf{p} - \lambda \mathbf{r})^2 + m^2}} \left[\frac{m^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (390)$$

The proper order of limits is $\lambda \ll \delta \implies \rho \rightarrow \infty$. The IR divergence is near the upper bound ρ (it has to be where λ is). Then

$$r\lambda \sim \rho\lambda = \frac{m}{\lambda} \delta\lambda = m\delta \rightarrow 0_+. \quad (391)$$

So I set λ to 0 in the integrand of Eq. 390. This changes the IR finite part, but not the IR divergence—which is what

I want.

$$\mathbb{T} = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr r^2 \frac{\delta[E_P - \sqrt{(\mathbf{P} + \mathbf{p})^2 + m^2}]}{\sqrt{r^2 + 1} \sqrt{(\mathbf{P} + \mathbf{p})^2 + m^2}} \left[\frac{m^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (392)$$

$$\mathbb{T} = \frac{\delta[E_P - \sqrt{(\mathbf{P} + \mathbf{p})^2 + m^2}]}{\sqrt{(\mathbf{P} + \mathbf{p})^2 + m^2}} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr \frac{r^2}{\sqrt{r^2 + 1}} \left[\frac{m^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (393)$$

In Eq. 274 I define $\mathcal{E} = \sqrt{(\mathbf{P} + \mathbf{p})^2 + m^2}$.

$$\mathbb{T} = \frac{\delta(E_P - \mathcal{E})}{\mathcal{E}} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr \frac{r^2}{\sqrt{r^2 + 1}} \left[\frac{m^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (394)$$

$$\mathbb{T} = \frac{\delta(E_P - \mathcal{E})}{\mathcal{E}} \Psi \quad (395)$$

$$\Psi = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr \frac{r^2}{\sqrt{r^2 + 1}} \left[\frac{m^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot \mathbf{P})[E_P \sqrt{r^2 + 1} - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (396)$$

I power count.

$$\Psi \sim \int_1^\rho dr \frac{r^2}{r^3} \left[\frac{m^2}{(E_P - |\mathbf{P}|)^2} + \frac{m^2}{(E_P - |\mathbf{p} + \mathbf{P}|)^2} + \frac{p^2}{(E_P - |\mathbf{P}|)(E_P - |\mathbf{p} + \mathbf{P}|)} \right] \quad (397)$$

$$\Psi \sim \left[\frac{m^2}{(E_P - |\mathbf{P}|)^2} + \frac{m^2}{(E_P - |\mathbf{p} + \mathbf{P}|)^2} + \frac{p^2}{(E_P - |\mathbf{P}|)(E_P - |\mathbf{p} + \mathbf{P}|)} \right] \int_1^\rho \frac{dr}{r} \quad (398)$$

$$\Psi \sim \int_1^\rho \frac{dr}{r} = \ln \rho \quad (399)$$

As I said: the IR divergence is at $r = \rho \rightarrow \infty$. To extract it, I expand the integrand for large r .

$$\Psi = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr \frac{r^2}{r} \left[\frac{m^2}{(E_P r - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P r - \mathbf{r} \cdot \mathbf{P})[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (400)$$

$$\Psi = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\rho dr r \left[\frac{m^2}{(E_P r - \mathbf{r} \cdot \mathbf{P})^2} + \frac{m^2}{[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} + \frac{p^2}{(E_P r - \mathbf{r} \cdot \mathbf{P})[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \right] \quad (401)$$

$$\Psi = \int_0^\rho dr r [m^2 \psi_1(r) + m^2 \psi_2(r) + p^2 \psi(r)] \quad (402)$$

$$\begin{aligned} \psi_1(r) &= \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P r - \mathbf{r} \cdot \mathbf{P})^2} \\ \psi_2(r) &= \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} \\ \psi(r) &= \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P r - \mathbf{r} \cdot \mathbf{P})[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \end{aligned} \quad (403)$$

$$\psi_1 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P r - \mathbf{r} \cdot \mathbf{P})^2} \quad (404)$$

$$\mathbf{P} = P_z \hat{\mathbf{z}} \quad (405)$$

$$\psi_1 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P r - r P_z \hat{\mathbf{r}} \cdot \hat{\mathbf{z}})^2} \quad (406)$$

$$r^2 \psi_1 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - P_z \hat{\mathbf{r}} \cdot \hat{\mathbf{z}})^2} \quad (407)$$

$$\hat{\mathbf{r}} \cdot \hat{\mathbf{z}} = \cos \theta \quad (408)$$

$$r^2 \psi_1 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - P_z \cos \theta)^2} \quad (409)$$

$$r^2 \psi_1 = 2\pi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - P_z \cos \theta)^2} \quad (410)$$

$$E_P = \sqrt{P_z^2 + m^2} \quad (411)$$

$$\int_0^\pi d\theta \frac{\sin \theta}{(a - b \cos \theta)^2} = \frac{2}{a^2 - b^2}, \quad 0 < |b| < |a| \quad (412)$$

$$0 < P_z < E_P = \sqrt{P_z^2 + m^2} \quad (413)$$

$$r^2 \psi_1 = 2\pi \frac{2}{E_P^2 - P_z^2} \quad (414)$$

$$\psi_1 = \frac{4\pi}{m^2 r^2} \quad (415)$$

$$\psi_2 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]^2} \quad (416)$$

$$r^2 \psi_2 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{[E_P - \hat{\mathbf{r}} \cdot (\mathbf{p} + \mathbf{P})]^2} \quad (417)$$

This integral is invariant under rotations: I orient the z -axis so that $\mathbf{p} + \mathbf{P} = |\mathbf{p} + \mathbf{P}| \hat{\mathbf{z}}$.

$$r^2 \psi_2 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - |\mathbf{p} + \mathbf{P}| \hat{\mathbf{r}} \cdot \hat{\mathbf{z}})^2} \quad (418)$$

$$r^2 \psi_2 = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - |\mathbf{p} + \mathbf{P}| \cos \theta)^2} \quad (419)$$

The Dirac delta function in Eq. 395, $\delta(E_P - \mathcal{E})$, implies $E_P = \mathcal{E}$.

$$r^2 \psi_2 = 2\pi \int_0^\pi d\theta \frac{\sin \theta}{(\mathcal{E} - |\mathbf{p} + \mathbf{P}| \cos \theta)^2} \quad (420)$$

$$0 < |\mathbf{p} + \mathbf{P}| < \mathcal{E} = \sqrt{(\mathbf{p} + \mathbf{P})^2 + m^2} \quad (421)$$

$$r^2 \psi_2 = \frac{4\pi}{\mathcal{E}^2 - (\mathbf{p} + \mathbf{P})^2} \quad (422)$$

$$\psi_2 = \frac{4\pi}{m^2 r^2} \quad (423)$$

$$\psi(r) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P r - \mathbf{r} \cdot \mathbf{P})[E_P r - \mathbf{r} \cdot (\mathbf{p} + \mathbf{P})]} \quad (424)$$

$$\psi(r) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P r - r \hat{\mathbf{r}} \cdot \mathbf{P})[E_P r - r \hat{\mathbf{r}} \cdot (\mathbf{p} + \mathbf{P})]} \quad (425)$$

$$r^2 \psi(r) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - \hat{\mathbf{r}} \cdot \mathbf{P})[E_P - \hat{\mathbf{r}} \cdot (\mathbf{p} + \mathbf{P})]} \quad (426)$$

I introduce Feynman parameters.

$$A = E_P - \hat{\mathbf{r}} \cdot \mathbf{P}, \quad B = E_P - \hat{\mathbf{r}} \cdot (\mathbf{p} + \mathbf{P}) \quad (427)$$

$$r^2 \psi(r) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{AB} \quad (428)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[A + (B - A)x]^2} \quad (429)$$

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[E_P - \hat{\mathbf{r}} \cdot (\mathbf{P} + x\mathbf{p})]^2} \quad (430)$$

$$r^2 \psi = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^1 \frac{dx}{[E_P - \hat{\mathbf{r}} \cdot (\mathbf{P} + x\mathbf{p})]^2} \quad (431)$$

$$r^2 \psi = \int_0^1 dx \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{[E_P - \hat{\mathbf{r}} \cdot (\mathbf{P} + x\mathbf{p})]^2} \quad (432)$$

I orient the z -axis along $\mathbf{P} + x\mathbf{p}$.

$$r^2 \psi = \int_0^1 dx \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - \hat{\mathbf{r}} \cdot |\mathbf{P} + x\mathbf{p}| \hat{\mathbf{z}})^2} \quad (433)$$

$$r^2 \psi = \int_0^1 dx \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - |\mathbf{P} + x\mathbf{p}| \hat{\mathbf{r}} \cdot \hat{\mathbf{z}})^2} \quad (434)$$

$$r^2 \psi = \int_0^1 dx \int_0^{2\pi} d\phi \int_0^\pi d\theta \frac{\sin \theta}{(E_P - |\mathbf{P} + x\mathbf{p}| \cos \theta)^2} \quad (435)$$

$$r^2 \psi = 2\pi \int_0^1 dx \int_0^\pi d\theta \frac{\sin \theta}{(E_P - |\mathbf{P} + x\mathbf{p}| \cos \theta)^2} \quad (436)$$

I show that

$$|\mathbf{P} + x\mathbf{p}| < E_P. \quad (437)$$

$$|\mathbf{P} + x\mathbf{p}|^2 = \mathbf{P}^2 + x^2 \mathbf{p}^2 + 2x \mathbf{P} \cdot \mathbf{p} < E_P^2 \quad (438)$$

$$E_P^2 = \mathcal{E}^2 \implies \mathbf{P}^2 = (\mathbf{P} + \mathbf{p})^2 \implies 2\mathbf{P} \cdot \mathbf{p} = -\mathbf{p}^2 \quad (439)$$

$$\mathbf{P}^2 + x^2 \mathbf{p}^2 - x \mathbf{p}^2 < E_P^2 \quad (440)$$

$$\mathbf{P}^2 - x(1-x)\mathbf{p}^2 < E_P^2 \quad (441)$$

In the range $0 \leq x \leq 1$

$$\mathbf{P}^2 - |x(1-x)|\mathbf{p}^2 < E_P^2 \quad (442)$$

$$\mathbf{P}^2 < E_P^2 + |x(1-x)|\mathbf{p}^2 \quad (443)$$

$$\mathbf{P}^2 < \mathbf{P}^2 + m^2 + |x(1-x)|\mathbf{p}^2 \quad (444)$$

$$0 < m^2 + |x(1-x)|\mathbf{p}^2 \quad (445)$$

This is true so $0 < |\mathbf{P} + x\mathbf{p}| < E_P$ and I can use Eq. 412 to integrate Eq. 436.

$$r^2 \psi = 4\pi \int_0^1 \frac{dx}{E_P^2 - (\mathbf{P} + x\mathbf{p})^2} \quad (446)$$

$$r^2 \psi = 4\pi \int_0^1 \frac{dx}{\mathbf{P}^2 + m^2 - (\mathbf{P}^2 + x^2 \mathbf{p}^2 + 2x \mathbf{P} \cdot \mathbf{p})} \quad (447)$$

$$2\mathbf{P} \cdot \mathbf{p} = -\mathbf{p}^2 \quad (448)$$

$$r^2 \psi = 4\pi \int_0^1 \frac{dx}{\mathbf{P}^2 + m^2 - (\mathbf{P}^2 + x^2 \mathbf{p}^2 - x \mathbf{p}^2)} \quad (449)$$

$$r^2 \psi = 4\pi \int_0^1 \frac{dx}{m^2 - x^2 \mathbf{p}^2 + x \mathbf{p}^2} \quad (450)$$

$$\psi = \frac{4\pi}{m^2 r^2} \int_0^1 \frac{dx}{1 + x(1-x) \frac{\mathbf{p}^2}{m^2}} \quad (451)$$

$$\int_0^1 \frac{dx}{1 + x(1-x)a} = \frac{2}{\sqrt{(4+a)a}} \ln \left[\frac{2}{2 + a - \sqrt{(4+a)a}} \right], \quad a > 0 \quad (452)$$

$$\psi = \frac{8\pi}{m^2 r^2 \sqrt{(4 + \frac{\mathbf{p}^2}{m^2}) \frac{\mathbf{p}^2}{m^2}}} \ln \left[\frac{2}{2 + \frac{\mathbf{p}^2}{m^2} - \sqrt{(4 + \frac{\mathbf{p}^2}{m^2}) \frac{\mathbf{p}^2}{m^2}}} \right] \quad (453)$$

$$\psi = \frac{8\pi}{mr^2 |\mathbf{p}| \sqrt{\frac{\mathbf{p}^2}{m^2} + 4}} \ln \left(\frac{2}{2 + \frac{\mathbf{p}^2}{m^2} - \frac{|\mathbf{p}|}{m} \sqrt{\frac{\mathbf{p}^2}{m^2} + 4}} \right) \quad (454)$$

$$\psi = \frac{8\pi}{r^2 \mathbf{p}^2} \ln \left(\frac{2}{2 + \frac{\mathbf{p}^2}{m^2} - \frac{\mathbf{p}^2}{m^2} \sqrt{1 + 4 \frac{m^2}{\mathbf{p}^2}}} \right) \quad (455)$$

$$p^2 = \vec{p}^2 = (0, \mathbf{p})^2 = -\mathbf{p}^2 \implies \mathbf{p}^2 = -p^2 \quad (456)$$

I take $-p^2/m^2 = \mathbf{p}^2/m^2 \rightarrow \infty$.

$$\psi = \frac{8\pi}{r^2 \mathbf{p}^2} \ln \left[\frac{2}{2 + \frac{\mathbf{p}^2}{m^2} - \frac{\mathbf{p}^2}{m^2} \left(1 + 2 \frac{m^2}{\mathbf{p}^2} - 2 \frac{m^4}{\mathbf{p}^4} \right)} \right] \quad (457)$$

$$\psi = \frac{8\pi}{r^2 \mathbf{p}^2} \ln \left(\frac{2}{2 - 2 + 2 \frac{m^2}{\mathbf{p}^2}} \right) = \frac{8\pi}{r^2 \mathbf{p}^2} \ln \left(\frac{1}{1 - 1 + \frac{m^2}{\mathbf{p}^2}} \right) \quad (458)$$

$$\psi = \frac{8\pi}{r^2 \mathbf{p}^2} \ln \left(\frac{\mathbf{p}^2}{m^2} \right) \quad (459)$$

$$\psi = -\frac{8\pi}{r^2 p^2} \ln \left(-\frac{p^2}{m^2} \right) \quad (460)$$

$$\psi_1 = \psi_2 = \frac{4\pi}{m^2 r^2} \quad (461)$$

$$\Psi = \int_0^\rho dr r (m^2 \psi_1 + m^2 \psi_2 + p^2 \psi) \quad (462)$$

$$\Psi = 8\pi \left[1 - \ln \left(-\frac{p^2}{m^2} \right) \right] \int_0^\rho dr \frac{r}{r^2} \quad (463)$$

$$\Psi = 8\pi \left[1 - \ln \left(-\frac{p^2}{m^2} \right) \right] \int_0^\rho \frac{dr}{r} \quad (464)$$

I take $-p^2/m^2 \rightarrow \infty$.

$$\Psi = -8\pi \ln \left(-\frac{p^2}{m^2} \right) \int_1^\rho \frac{dr}{r} \quad (465)$$

I only want the IR divergence (upper bound), and have consistently ignored IR finite terms, so I move the lower bound up: the apparent $\ln 0$ divergence comes from ignoring IR finite terms.

$$\Psi = -8\pi \ln \left(-\frac{p^2}{m^2} \right) \ln \rho, \quad \rho = \frac{m}{\lambda} \delta \quad (466)$$

$$\Psi = -8\pi \ln \left(-\frac{p^2}{m^2} \right) \left[\ln \left(\frac{m}{\lambda} \right) + \ln \delta \right] \quad (467)$$

I justify ignoring the $\ln \delta$ term in Sec. 3.3.7.

$$\boxed{\Psi = -8\pi \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{m}{\lambda} \right)} \quad (468)$$

8.6. Decay Rate

$$\Psi = -8\pi \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{m}{\lambda} \right) \quad (469)$$

$$\mathbb{T} = \frac{\delta(E_P - \mathcal{E})}{\mathcal{E}} \Psi \quad (470)$$

$$\mathbb{T} = -8\pi \frac{\delta(E_P - \mathcal{E})}{\mathcal{E}} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{m}{\lambda} \right) \quad (471)$$

$$\Gamma_B = -\frac{e^2 |\mathcal{M}_0|^2 \mathbb{T}}{32\pi^2 E_P} \quad (472)$$

$$\Gamma_B = \frac{e^2 |\mathcal{M}_0|^2}{4\pi E_P \mathcal{E}} \delta(E_P - \mathcal{E}) \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{m}{\lambda} \right) \quad (473)$$

$$\Gamma_0 = \frac{\pi |\mathcal{M}_0|^2}{2E_P \mathcal{E}} \delta(E_P - \mathcal{E}) \quad (474)$$

$$\frac{2}{\pi} \Gamma_0 = \frac{|\mathcal{M}_0|^2}{E_P \mathcal{E}} \delta(E_P - \mathcal{E}) \quad (475)$$

$$\boxed{\Gamma_B = \frac{e^2 \Gamma_0}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{m}{\lambda} \right)} \quad (476)$$

9. TOTAL DECAY RATE

I drop all IR finite terms. The $1 \rightarrow 1$ decay rate (Eq. 287) is

$$\Gamma = \frac{e^2}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{\lambda}{m} \right) \Gamma_0. \quad (477)$$

The bremsstrahlung decay rate (Eq. 476) is

$$\Gamma_B = \frac{e^2 \Gamma_0}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln \left(\frac{m}{\lambda} \right). \quad (478)$$

The total decay rate is

$$\Gamma_T = \Gamma + \Gamma_B. \quad (479)$$

$$\Gamma_T = \frac{e^2 \Gamma_0}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \left[\ln \left(\frac{\lambda}{m} \right) + \ln \left(\frac{m}{\lambda} \right) \right] \quad (480)$$

$$\Gamma_T = \frac{e^2 \Gamma_0}{2\pi^2} \ln \left(-\frac{p^2}{m^2} \right) \ln 1 = 0 \quad (481)$$

The IR divergences cancel. QED

10. COMPARING SCALAR & SPINOR QED

Unlike regular QED, scalar QED has 4-way diagrams (Figs. 5, 6, & 12). But these do not contribute to the IR divergence. So the only difference, as far as the IR divergence goes, is that scalar QED is easier because there is no Dirac algebra.